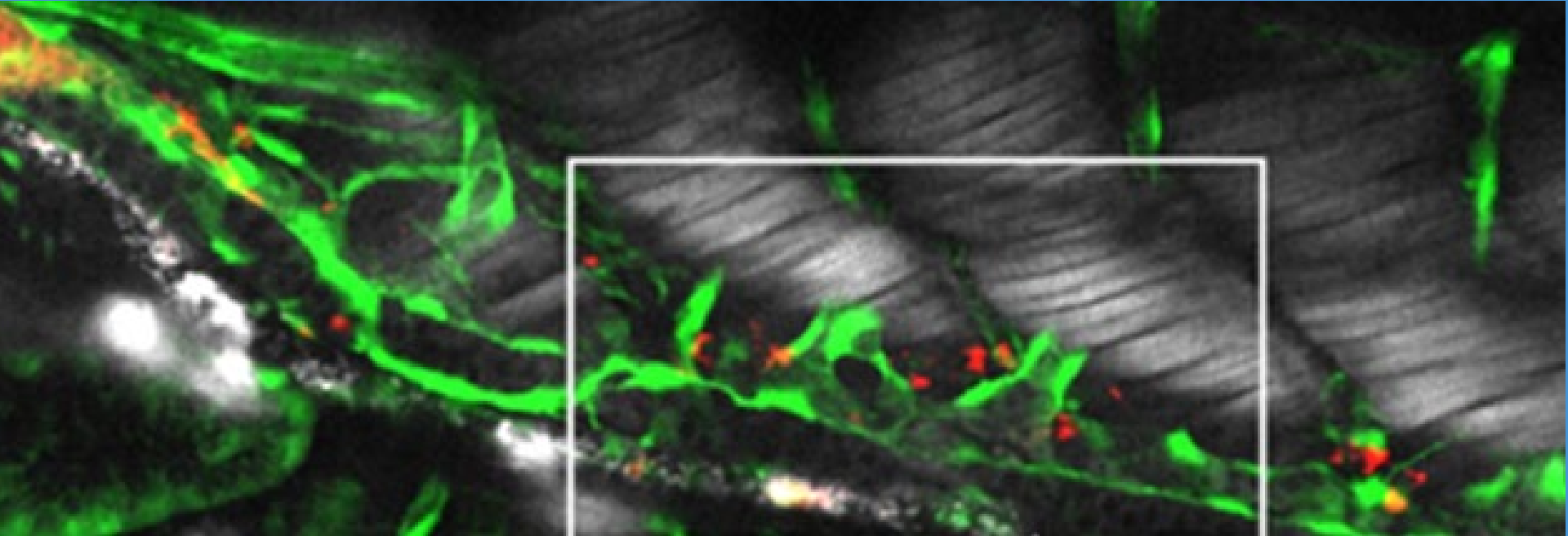




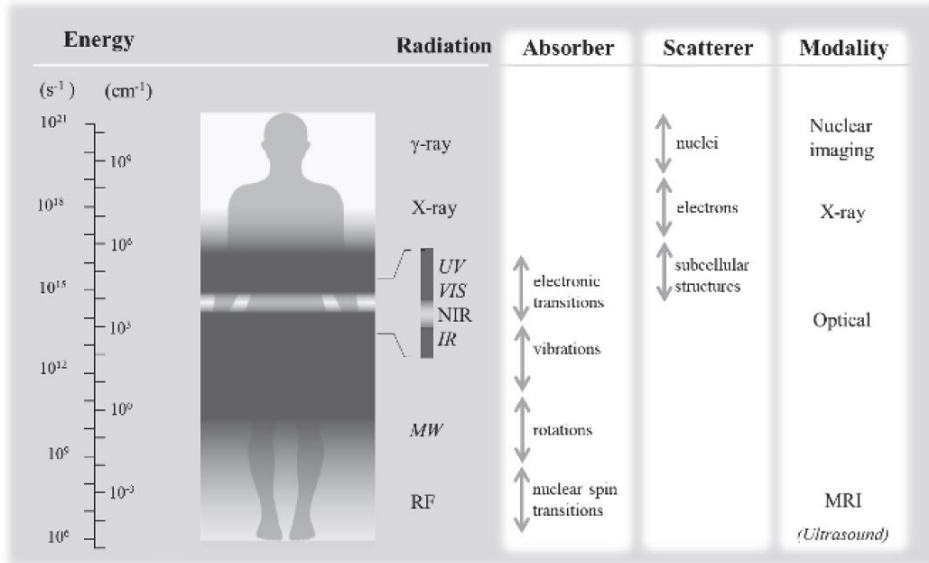
Optical *in vivo* imaging and microscopy

Imaging for the Life Sciences
Prof. Dr. Steffi Lehmann



Topic of today: optical imaging... Relevance?

What do we see with visible light?



Not much... Less than 1mm...



Topic of today: optical imaging... Relevance?



Light in the
visible domain of
the spectrum is

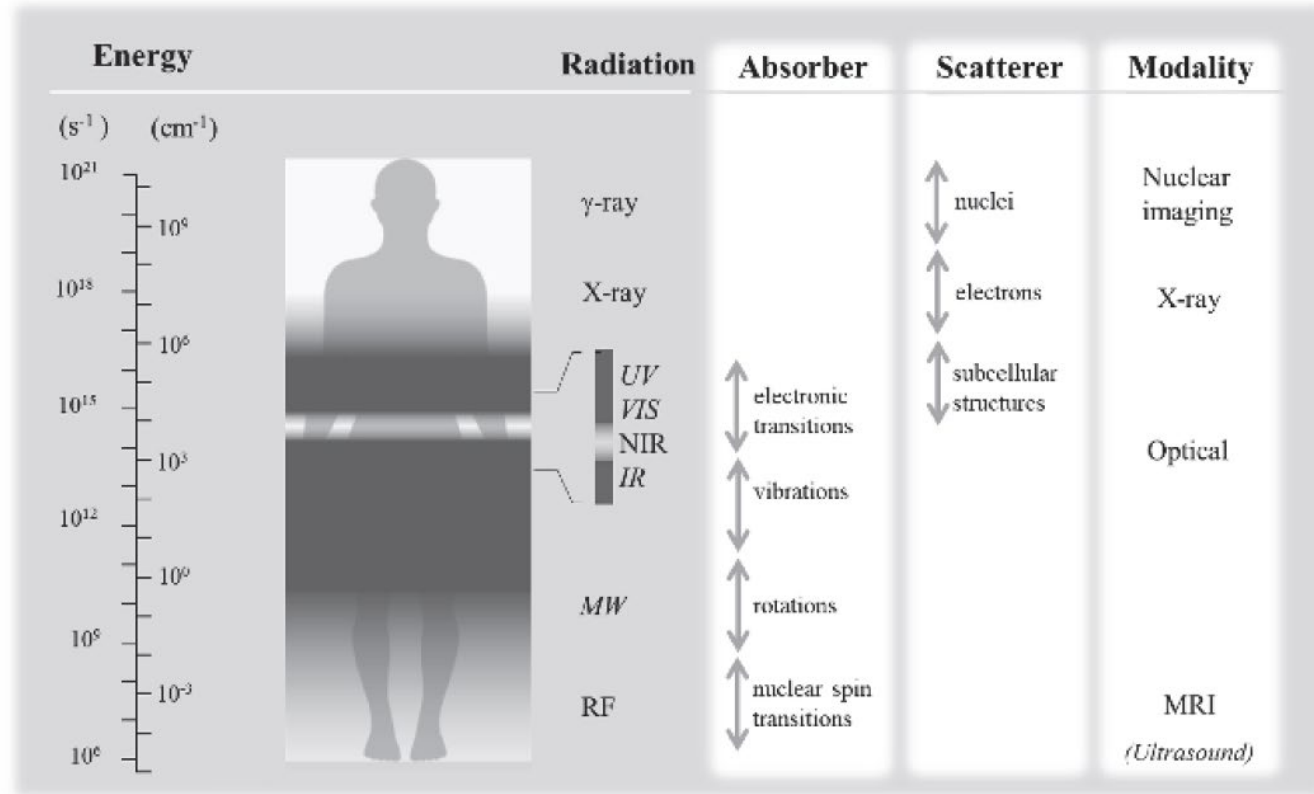
→ **absorbed**

→ **scattered**

**“There is a crack in everything,
that's how the light gets in.”**

Leonard Cohen

Biomedical imaging methods offer specific windows to the body



Markus Rudin, Molecular Imaging, Basic principles and applications in biomedical research, 3rd edition

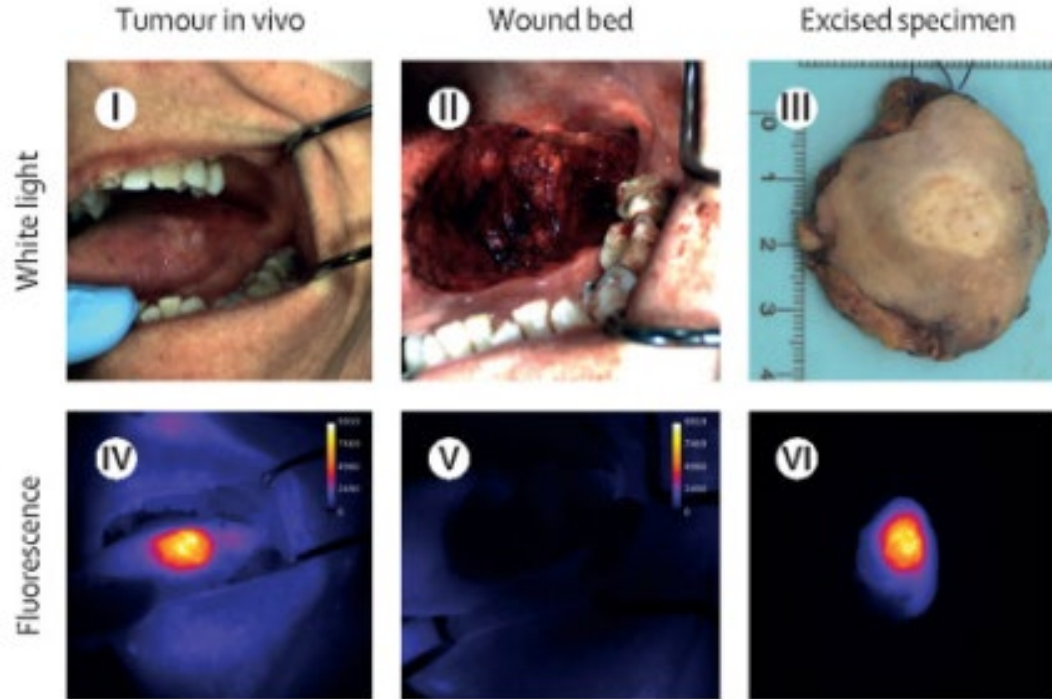
Topic of today: optical imaging... Relevance?

One important clinical application: **intraoperative fluorescence imaging to detect tumor margin.**

Fluorescence labeling of tumor cells using pH-sensitive probe.
Example of head and neck carcinoma in the tongue.

But there are more applications to be explored...

A Excision with tumour-free margins

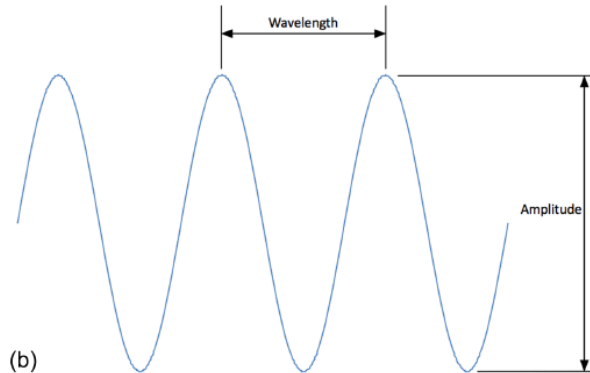
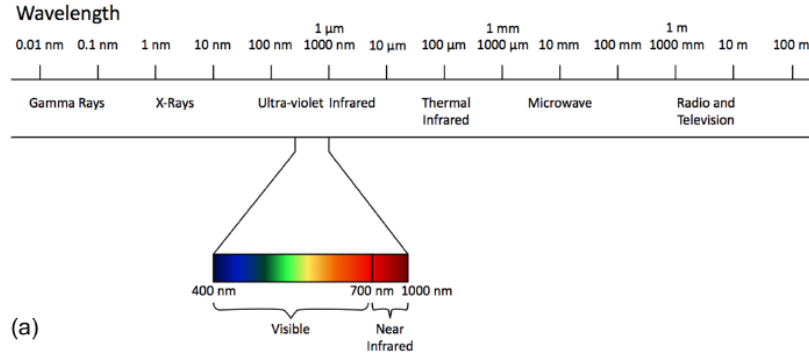


Program today

Introduction to **optical molecular imaging**:

- 1) **Principles of fluorescence**
- 2) **Imaging probes**: fluorescent small molecules and proteins
- 3) Imaging **equipment for macroscopic imaging**
- 4) Target specific- vs non-specific imaging
- 5) **Microscopic fluorescence** imaging
 - Intravital confocal microscopy
 - Intravital multiphoton microscope
- 6) Bioluminescence imaging
- 7) Wrap-Up

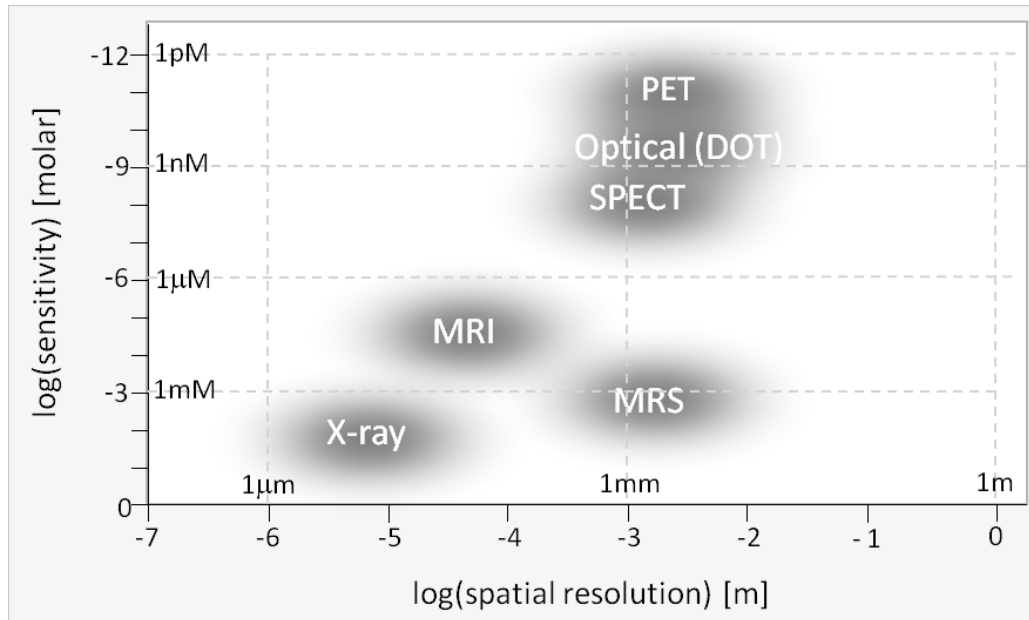
Optical imaging uses light to probe biological samples and generate contrast



Typically light in the visible or near infrared domain of the electromagnetic spectrum is used to generate contrast. As a wave, light has a defined wavelength and frequency. Light particles are photons. The higher the energy of photons, the lower the wavelength.

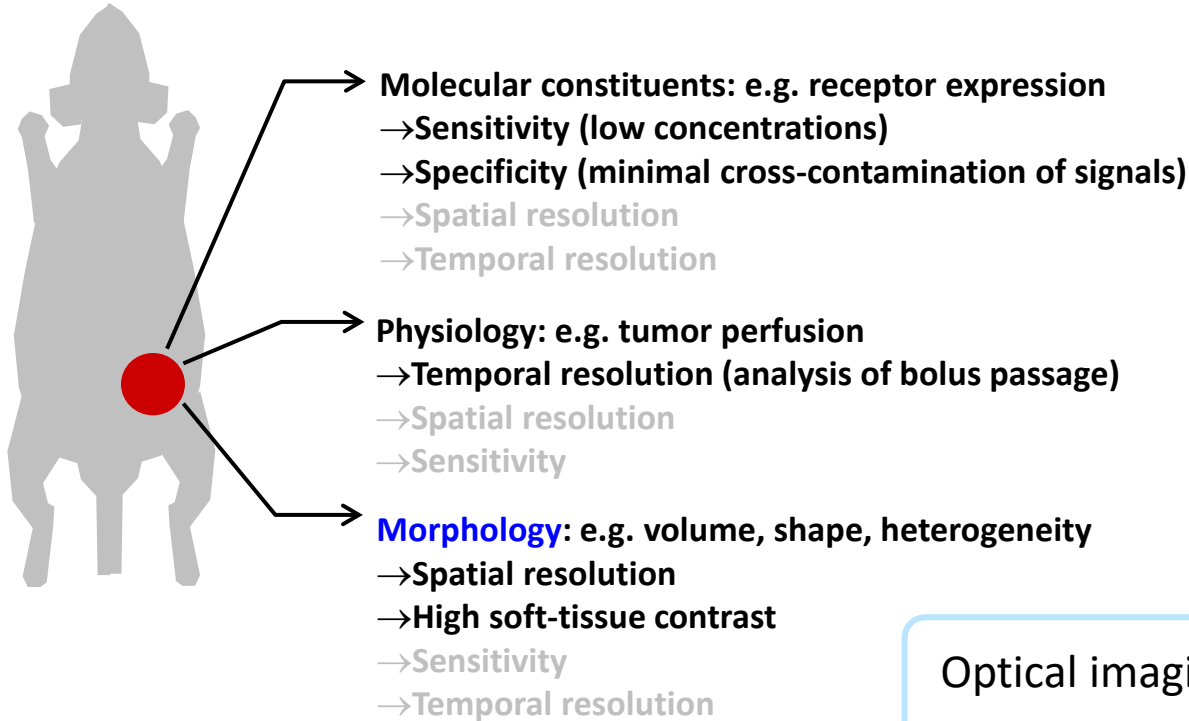
Comparison of the different imaging modalities

Spatial resolution versus sensitivity for the most important imaging methods



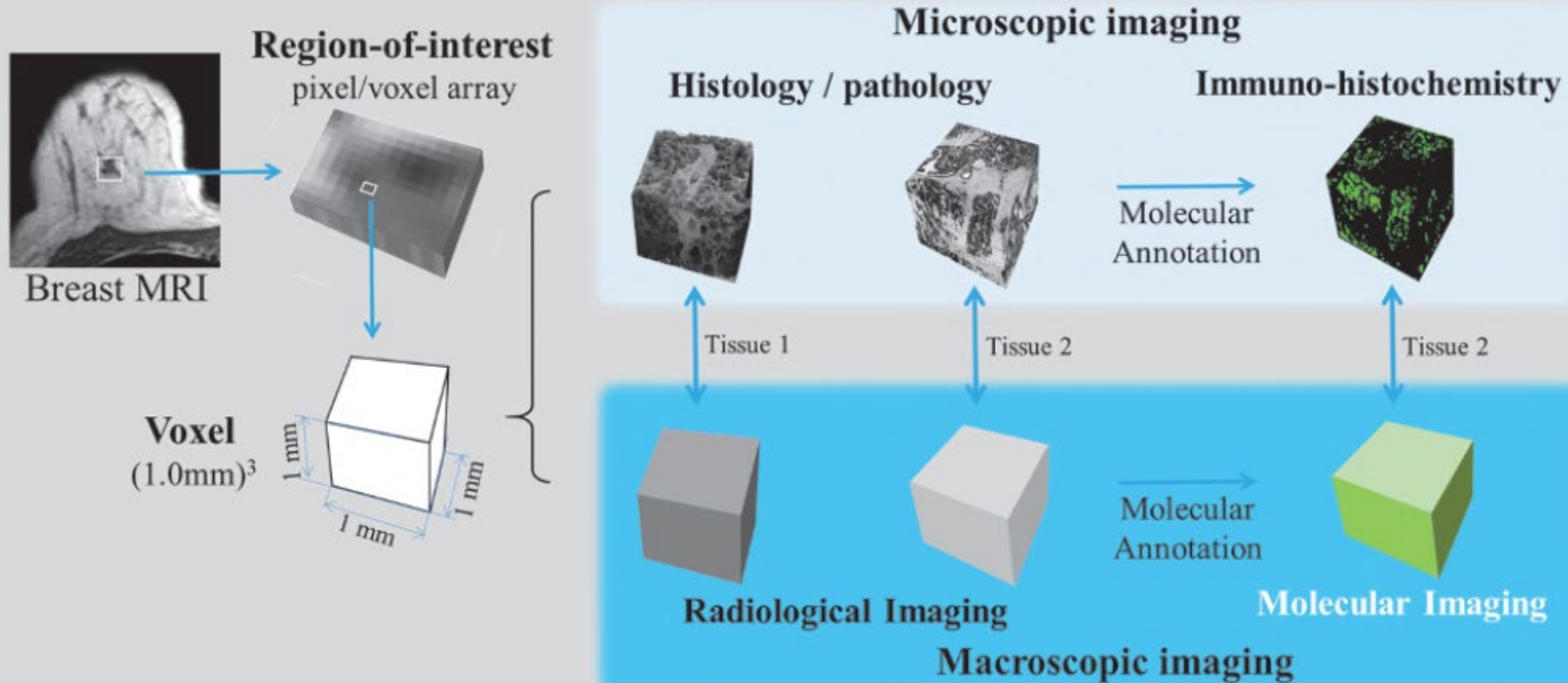
Non-invasive, *in vivo* optical imaging has poor spatial resolution. If one moves to intravital microscopy, this changes drastically...

Optimal imaging method depends on the question to be answered



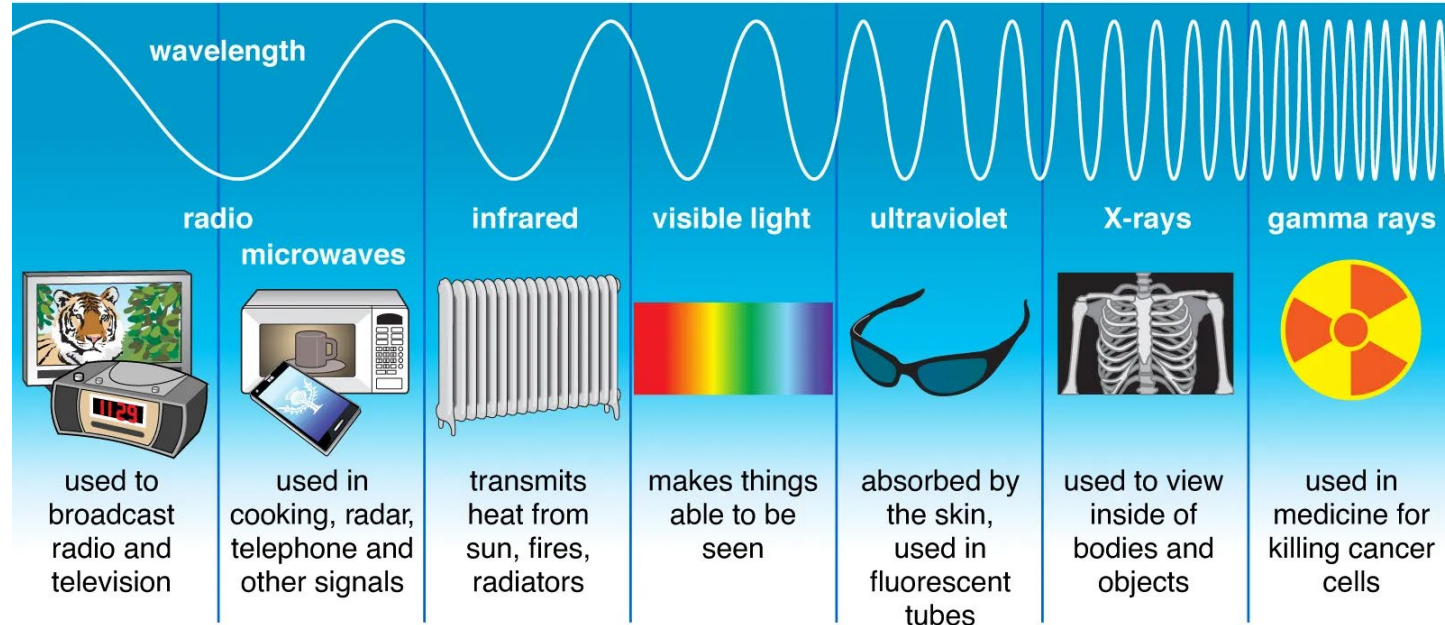
Optical imaging???

Optical imaging is particularly suited to obtain molecular information – at a macroscopic and microscopic level



Light in the visible and near infrared /infrared domain of the electromagnetic spectrum can be used for optical imaging

Types of Electromagnetic Radiation



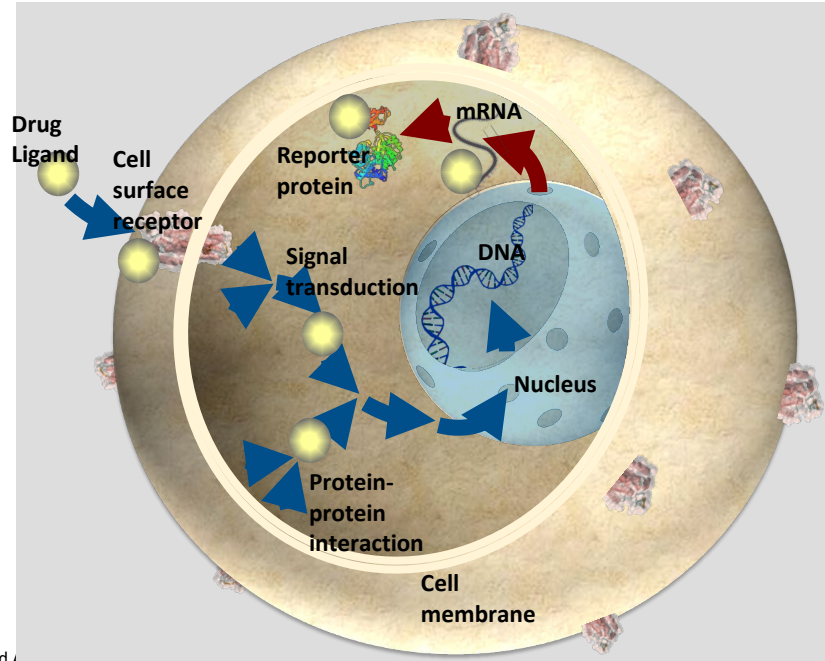
© Encyclopædia Britannica, Inc.

What can be imaged using optical imaging?

- Structural and anatomical information
- **Molecular information**

- Use of reporter proteins
- Use of exogenous contrast agents

Reporter probes are
fluorescent dyes or
fluorescent proteins

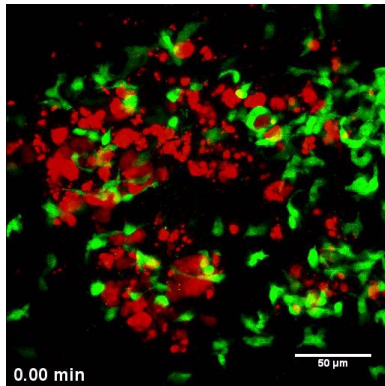


Program of today: use of optical imaging to gain *in vivo* molecular information

Fluorescence imaging

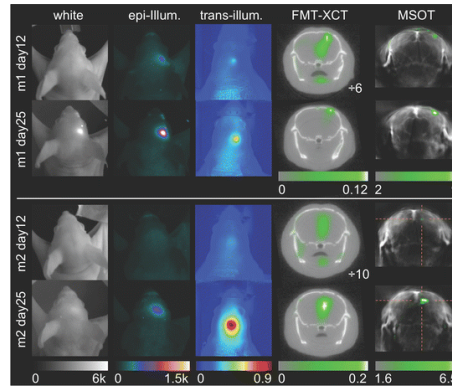
Intravital microscopy

Visualizing therapy induced killing of tumor cells by T cells



Fluorescence reflectance imaging or tomography

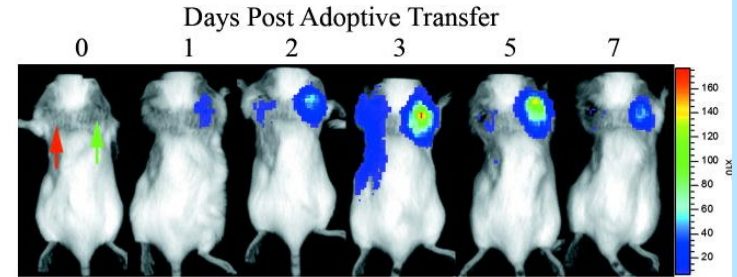
Monitoring tumor growth (glioblastoma in the mouse brain)



Deliolanis et al, Molecular Imaging and Biology, 2014

Bioluminescence imaging

Monitoring immune cell activation in a tumor model



Patel et al., Cancer Research, 2010

→ Contrast generated (mostly) using exogenous fluorescent and bioluminescent probes

Program today

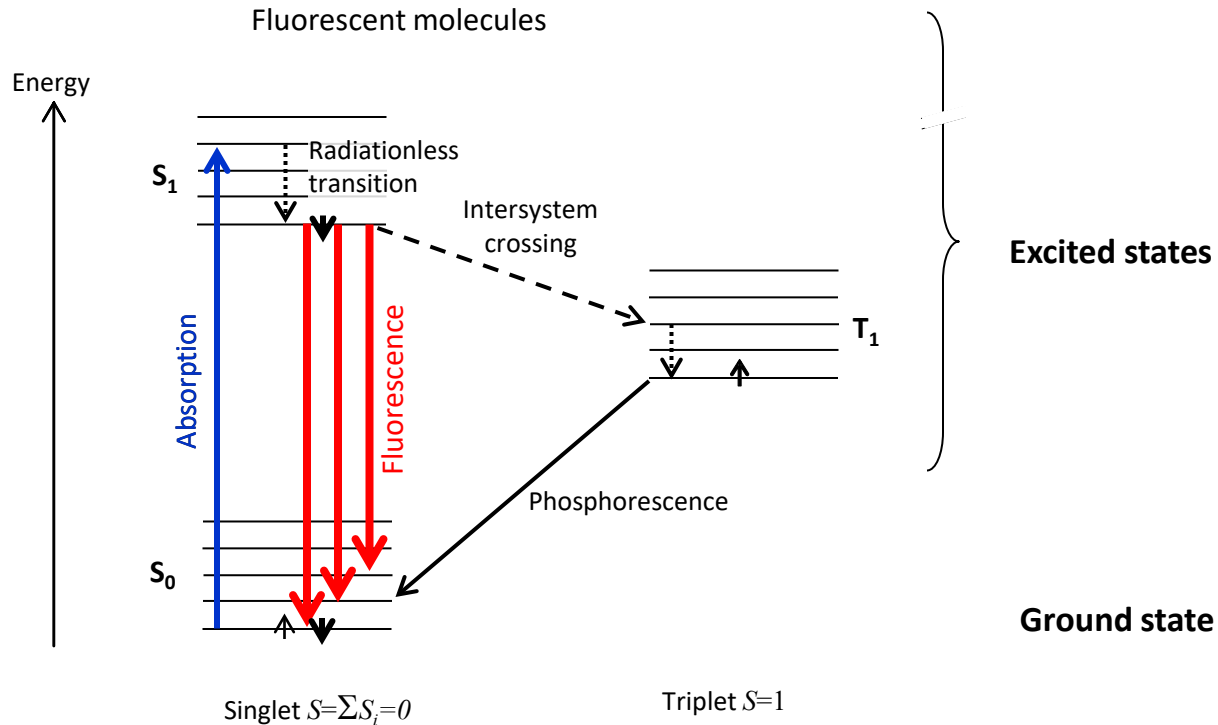
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Optical imaging: Principles of fluorescence

«Emission of light photons due to transitions from an electronically excited state into the ground state»

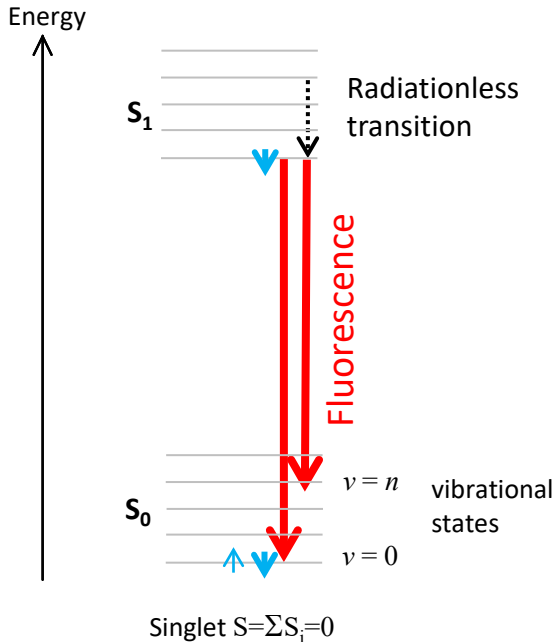
Energy level diagram – Jablonski diagram



Optical imaging: Principles of fluorescence imaging

system relaxes

- to the vibrational ground state of S_1 by radiationless energy dissipation
- to the electronic ground-state S_0 by emission of a photon of the energy



Emission / fluorescence:

$$\hbar \cdot \omega_{fl} = E(S_1, v=0) - E(S_0, v=n)$$

$\omega_{fl} < \omega_0$ i.e. fluorescence occurs always at longer wavelengths than the excitation

-> Stokes shift

Characteristic times for absorption & emission processes

Excitation	$S_{0,v_0} \rightarrow S_n,v_j$	10^{-15}s
Internal conversion	$S_1,v_j \rightarrow S_1,v_0$	$10^{-11}\text{-}10^{-14}\text{s}$
Fluorescence	$S_1,v_0 \rightarrow S_0,v_i + h\nu$	$10^{-7}\text{-}10^{-9}\text{s}$
Internal conversion	$S_1,v_0 \rightarrow S_0$	$10^{-5}\text{-}10^{-7}\text{s}$
Intersystem crossing	$S_1,v_0 \rightarrow T_1$	10^{-8}s
Phosphorescence	$T_1 \rightarrow S_0 + h\nu$	$1\text{-}10\text{s}$
Internal conversion	$T_1 \rightarrow S_0$	10^{-8}s

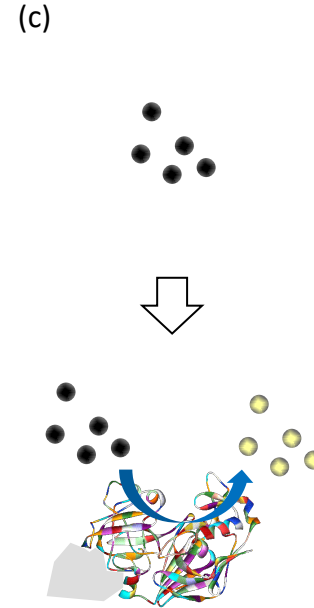
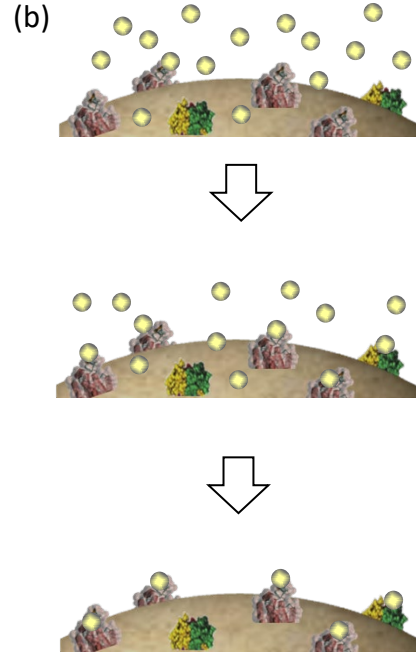
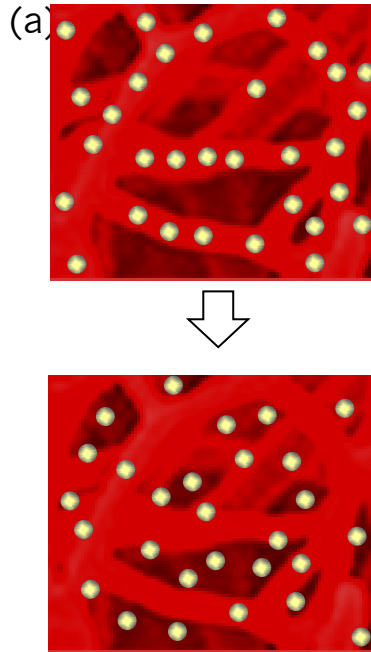
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Contrast agents for optical imaging are fluorescent molecules:

Non-specific versus target-specific contrast agents



Fluorescent probes and fluorescent/bioluminescent reporter proteins as contrast agents



Light absorption in visible / near infrared spectral domain

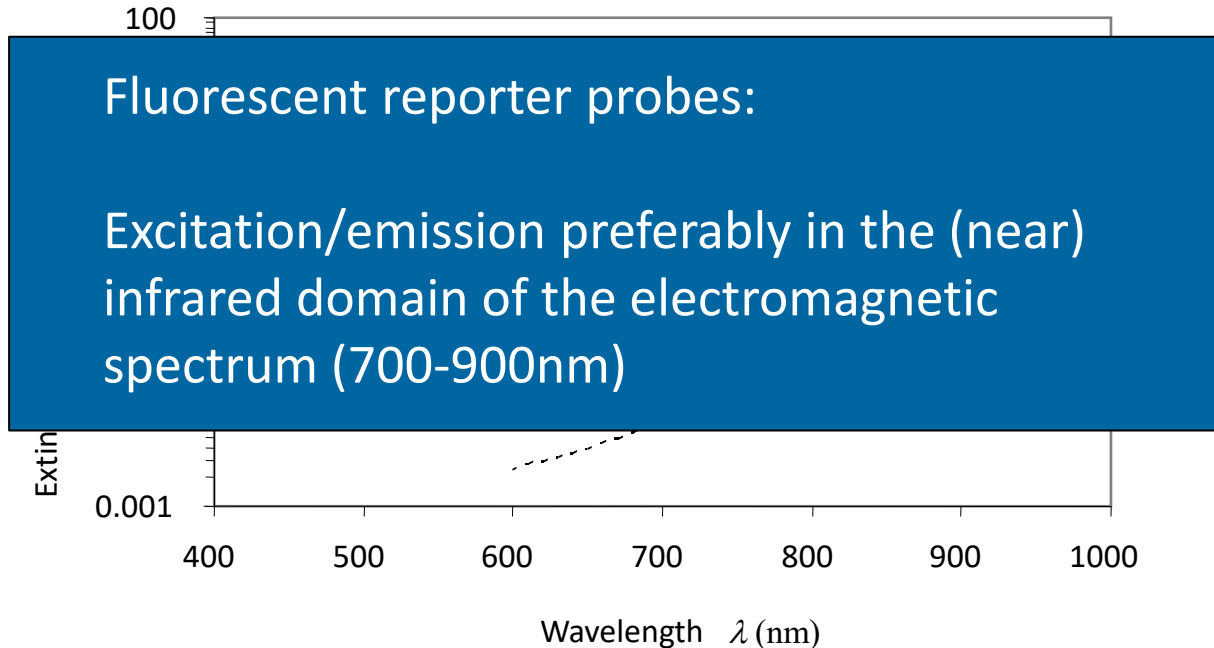
Beer-Lambert Law of Absorption

$$\frac{I(t)}{I_0} = \exp(-\varepsilon \cdot c(t) \cdot d)$$

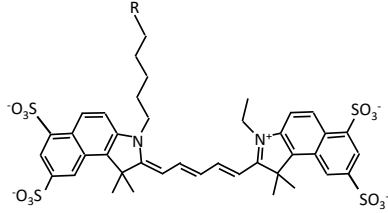
ε extinction/absorption coefficient

$c(t)$ concentration of absorber

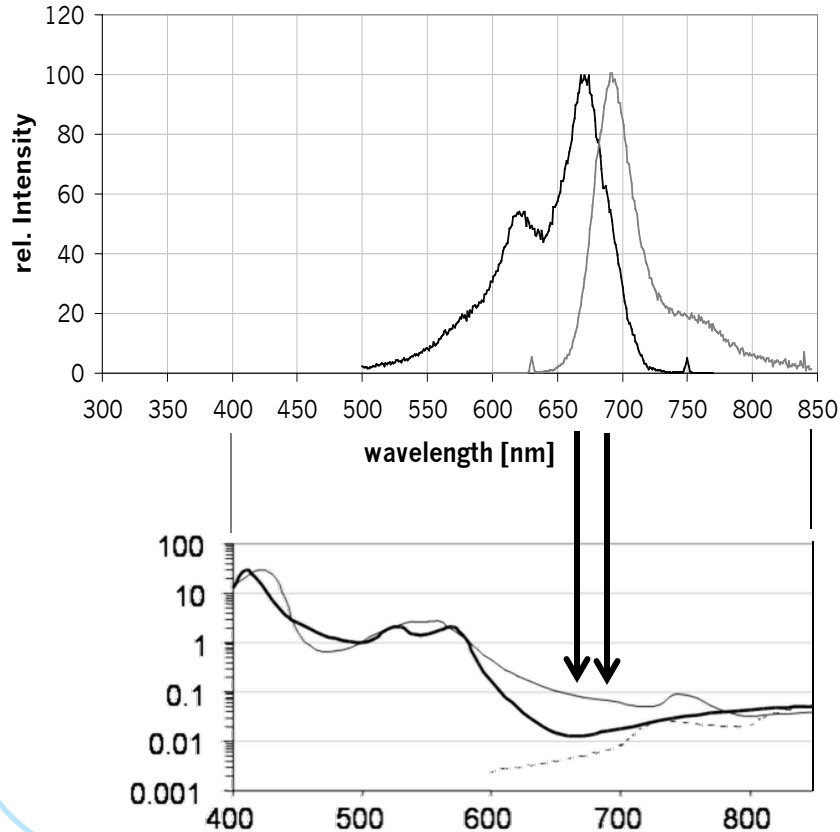
d path length of light



Absorption/emission spectrum of Cy5.5

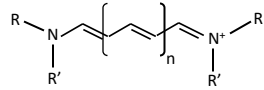


Cy5.5: a fluorescent dye typically used for in vivo imaging experiments

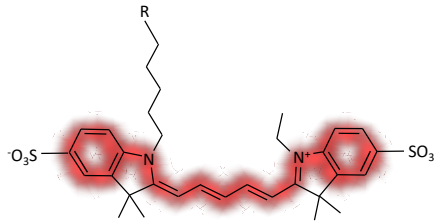


Near-infrared fluorescent dyes

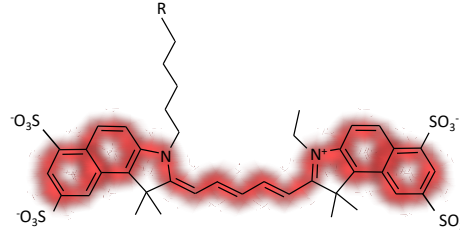
Basic structure of cyanine dyes



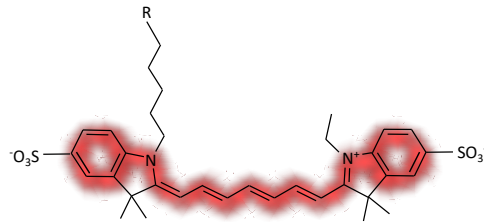
Indocyanine dyes



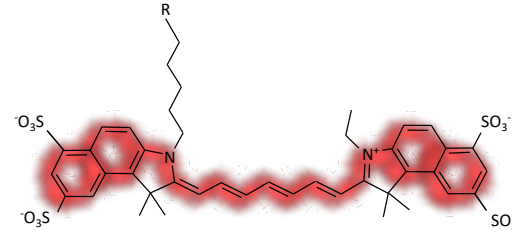
Cy5



Cy5.5 (=DBCy5)



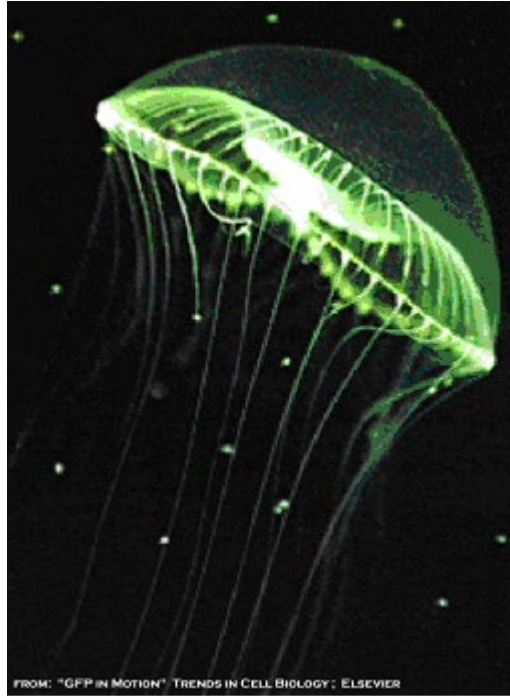
Cy7



DBCy7

Alternatively: fluorescent proteins

Green fluorescent protein (GFP)



Aequoria victoria

Nobel prize in Chemistry in 2008
Prof. Shimomura, Chalfie and Tsien

transgene
expression →

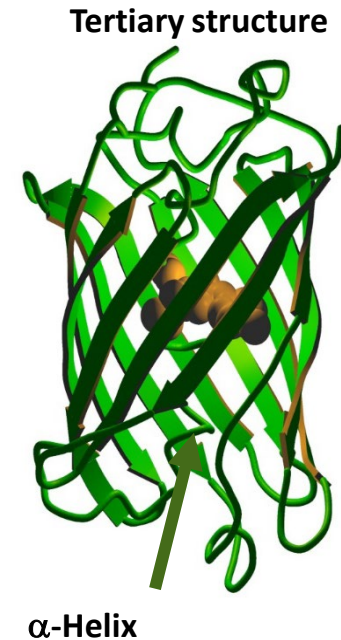
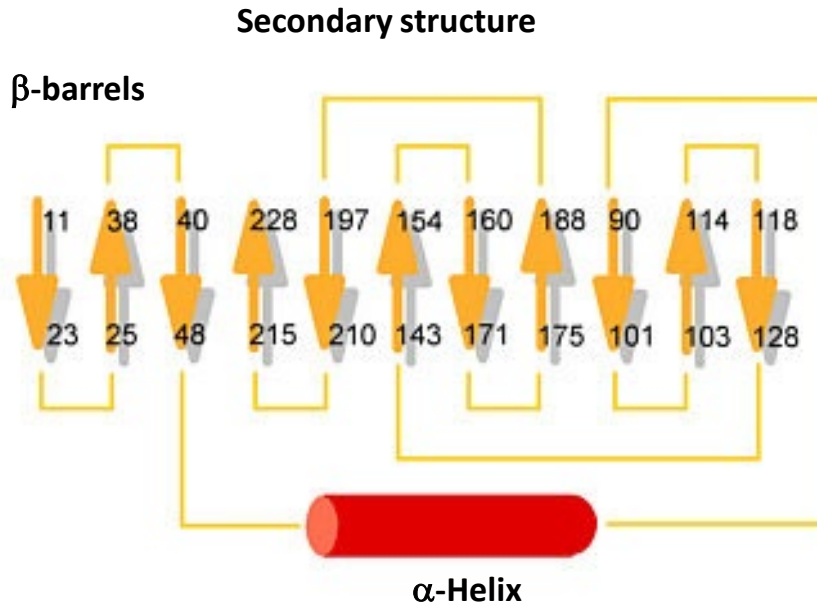


Drosophila melanogaster

→ Have the advantage that they can be used as genetically encoded reporters

Structure of GFP

- Barrel formed by β -sheet with slightly oval in cross-section.
- Distorted helix along the center of barrel with residues arranged in a helical pattern
- Structure inside barrel forms chromophore from amino acids Ser65, Tyr66 and Gly67 by an internal cyclization





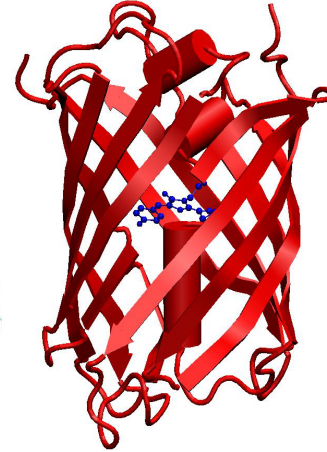
Red fluorescent proteins



DsRed from Discosoma



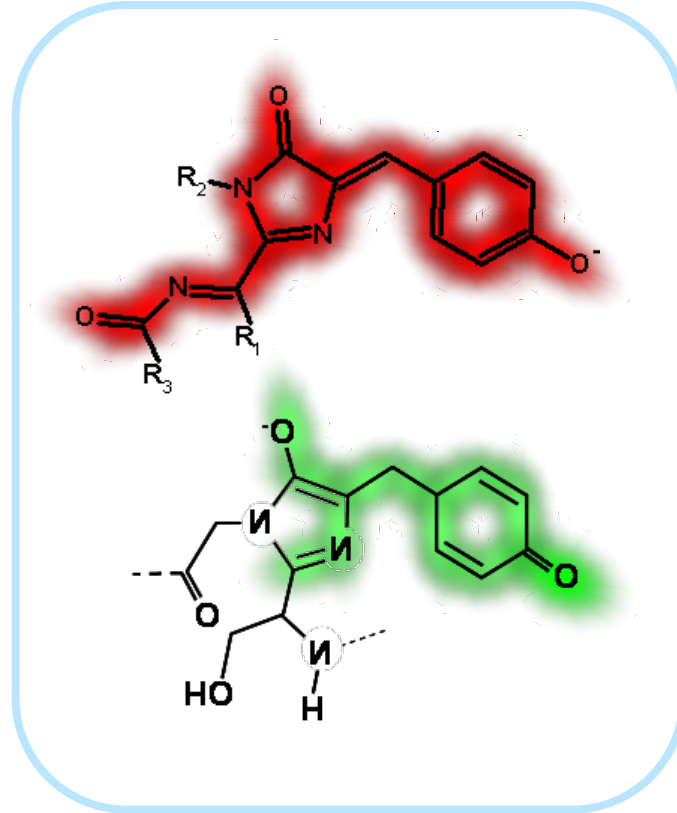
**Eqfp611 from
Enracmaea quadricolor**



**RFP from Discosoma,
derived from dsred**

Comparison of GFP and RFP Chromophore

Length of chromophor, i.e. of conjugated π electron system, determines color or more precisely wavelength of fluorescent light.



Fluorescent proteins

Table 1 | Properties of the best FP variants^{a,b}

Shaner, Steinbach, Tsien, *Nature Methods*, 2:905 (2005)

Class	Protein	Source laboratory (references)	Excitation ^c (nm)	Emission ^d (nm)	Brightness ^e	Photostability ^f	pKa	Oligomerization
Far-red	mPlum ^g	Tsien (5)	590	649	4.1	53	<4.5	Monomer
Red	mCherry ^g	Tsien (4)	587	610	16	96	<4.5	Monomer
	tdTomato ^g	Tsien (4)	554	581	95	98	4.7	Tandem dimer
	mStrawberry ^g	Tsien (4)	574	596	26	15	<4.5	Monomer
	J-Red ^h	Evrogen	584	610	8.8 [*]	13	5.0	Dimer
	DsRed-monomer ^h	Clontech	556	586	3.5	16	4.5	Monomer
Orange	mOrange ^g	Tsien (4)	548	562	49	9.0	6.5	Monomer
	mKO	MBL Intl. (10)	548	559	31 [*]	122	5.0	Monomer
Yellow-green	mCitrine ⁱ	Tsien (16,23)	516	529	59	49	5.7	Monomer
	Venus	Miyawaki (1)	515	528	53 [*]	15	6.0	Weak dimer ^j
	YPet ^g	Daugherty (2)	517	530	80 [*]	49	5.6	Weak dimer ^j
	EYFP	Invitrogen (18)	514	527	51	60	6.9	Weak dimer ^j
Green	Emerald ^g	Invitrogen (18)	487	509	39	0.69 ^k	6.0	Weak dimer ^j
	EGFP	Clontech ^l	488	507	34	174	6.0	Weak dimer ^j
Cyan	CyPet	Daugherty (2)	435	477	18 [*]	59	5.0	Weak dimer ^j
	mCFPm ^m	Tsien (23)	433	475	13	64	4.7	Monomer
	Cerulean ^g	Piston (3)	433	475	27 [*]	36	4.7	Weak dimer ^j
UV-excitable green	T-Sapphire ^g	Griesbeck (6)	399	511	26 [*]	25	4.9	Weak dimer ^j

^aAn expanded version of this table, including a list of other commercially available FPs, is available as **Supplementary Table 1**. ^bThe mutations of all common AFPs relative to the wild-type protein are available in **Supplementary Table 3**. ^cMajor excitation peak. ^dMajor emission peak. ^eProduct of extinction coefficient and quantum yield at pH 7.4 measured or confirmed (indicated by *) in our laboratory under ideal maturation conditions, in (mM * cm)⁻¹ (for comparison, free fluorescein at pH 7.4 has a brightness of about 69 (mM * cm)⁻¹). Time for bleaching from an initial emission rate of 1,000 photons/s down to 500 photons/s (1_{1/2}; for comparison, fluorescein at pH 8.4 has 1_{1/2} of 5.2 s); data are not indicative of photostability under focused laser illumination. ^fBrightest in spectral class. ^gNot recommended (dim with poor folding at 37 °C). ^hCitrine YFP with A206K mutation; spectroscopic properties equivalent to Citrine. ⁱCan be made monomeric with A206K mutation. ^jEmerald has a pronounced fast bleaching component that leads to a very short time to 50% bleach. Its photostability after the initial few seconds, however, is comparable to that of EGFP. ^kFormerly sold by Clontech, no longer commercially available. ^lmECFP with A206K mutation; spectroscopic properties equivalent to ECFP.

Green fluorescent proteins as reporters are still subject of active research

StayGold shows increased photostability

Brief Communication

<https://doi.org/10.1038/s41587-023-02018-w>

A monomeric StayGold fluorescent protein

Received: 12 March 2023

Accepted: 5 October 2023

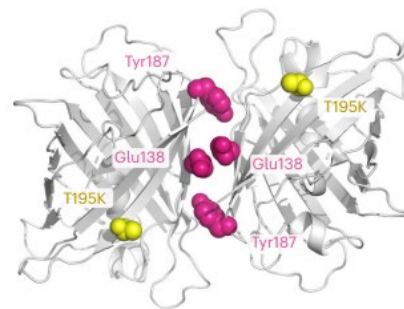
Published online: 11 December 2023

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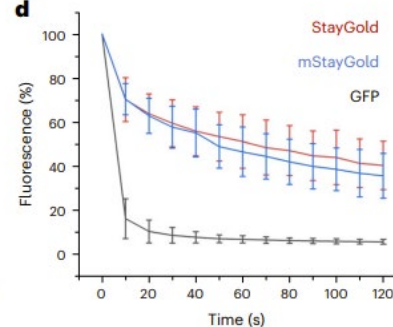
Esther Ivorra-Molla^{1,5}, Dipayan Akhuli^{1,2,5}, Martin B. L. McAndrew^{3,4}, William Scott¹, Lokesh Kumar¹, Saravanan Palani², Masanori Mishima¹✉, Allister Crow³✉ & Mohan K. Balasubramanian¹✉

StayGold is an exceptionally bright and stable fluorescent protein that is highly resistant to photobleaching. Despite favorable fluorescence properties, use of StayGold as a fluorescent tag is limited because it forms a natural dimer. Here we report the 1.6 Å structure of StayGold and generate a derivative, mStayGold, that retains the brightness and photostability of the original protein while being fully monomeric.

b



d

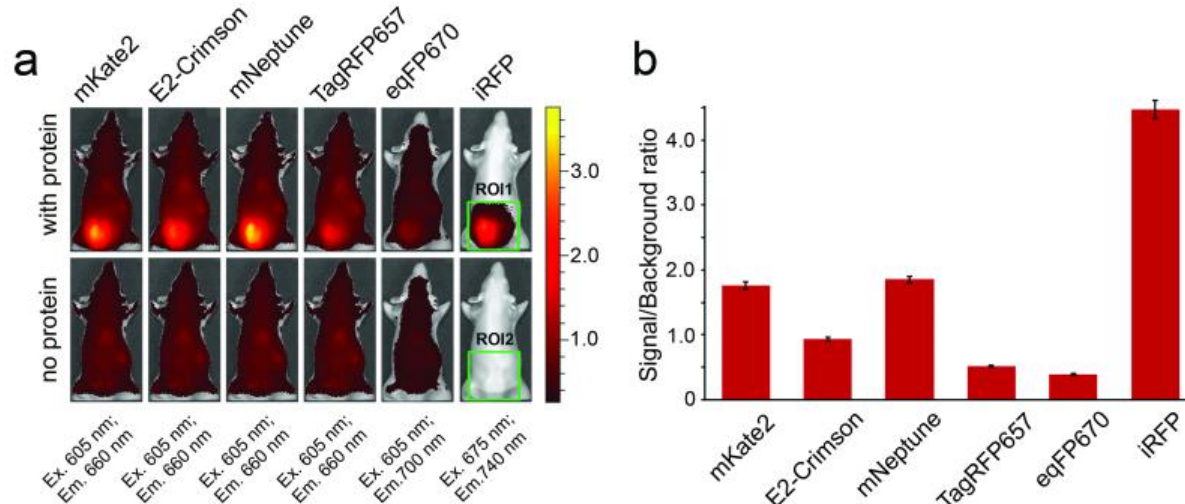


Fluorescent proteins - today?

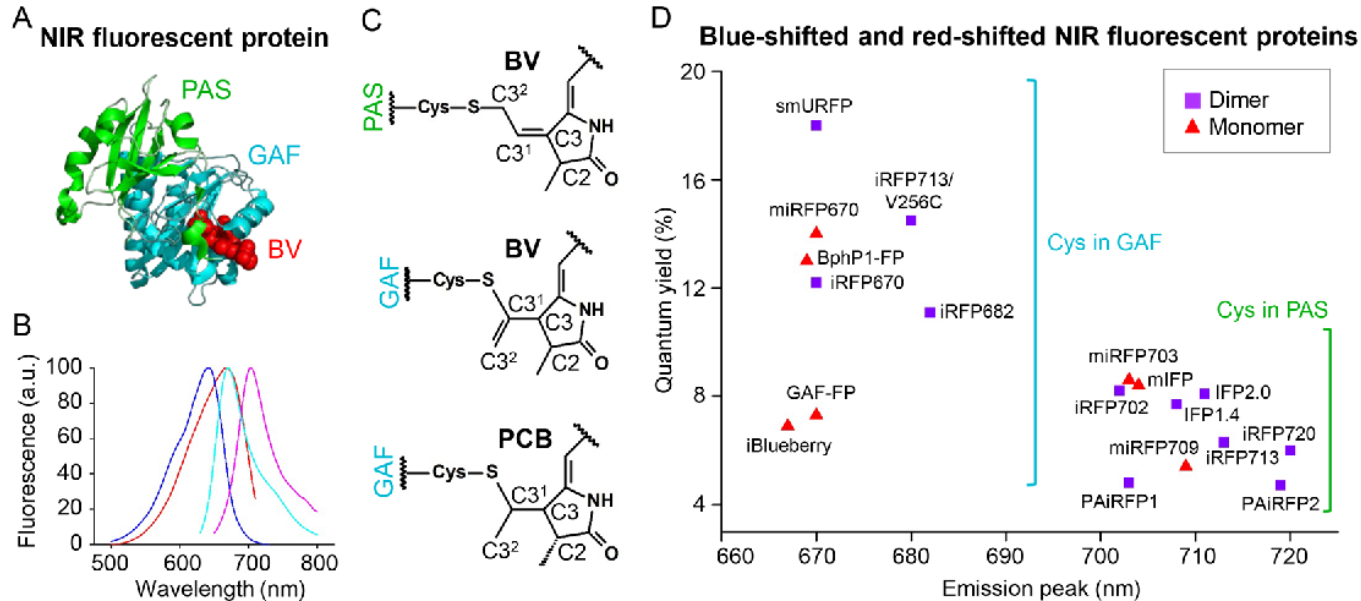
Novel fluorescence proteins based on bacteriophytochromes

Phytochromes = photosensory receptors absorbing light in the far red domain of the spectrum

→ Dependent on tetrapyrrole chromophore binding such as biliverdin (intermediate of the mammalian heme metabolism)



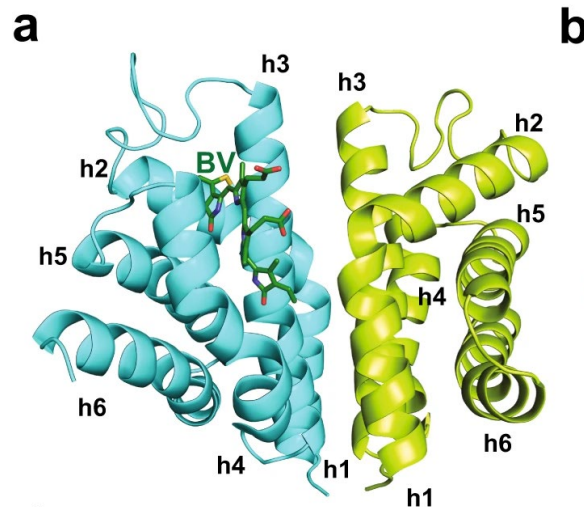
Bacterial phytochromes: true NIR fluorescent proteins



33



→ Attractive for in vivo imaging



Maiti, A., Buffalo, C.Z., Saurabh, S. et al. *Nat Commun* **14**, 4155 (2023).

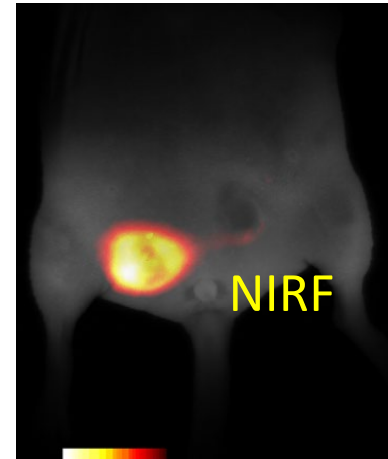
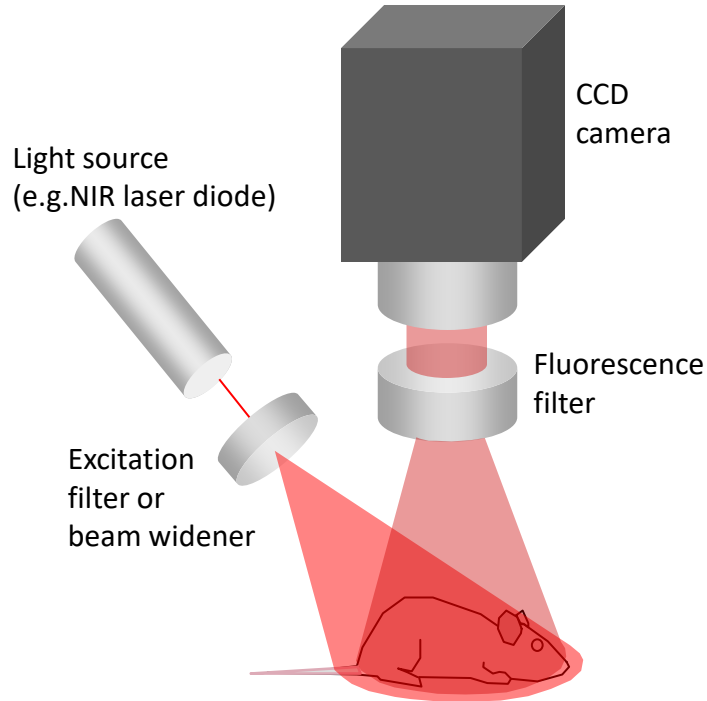


Program today

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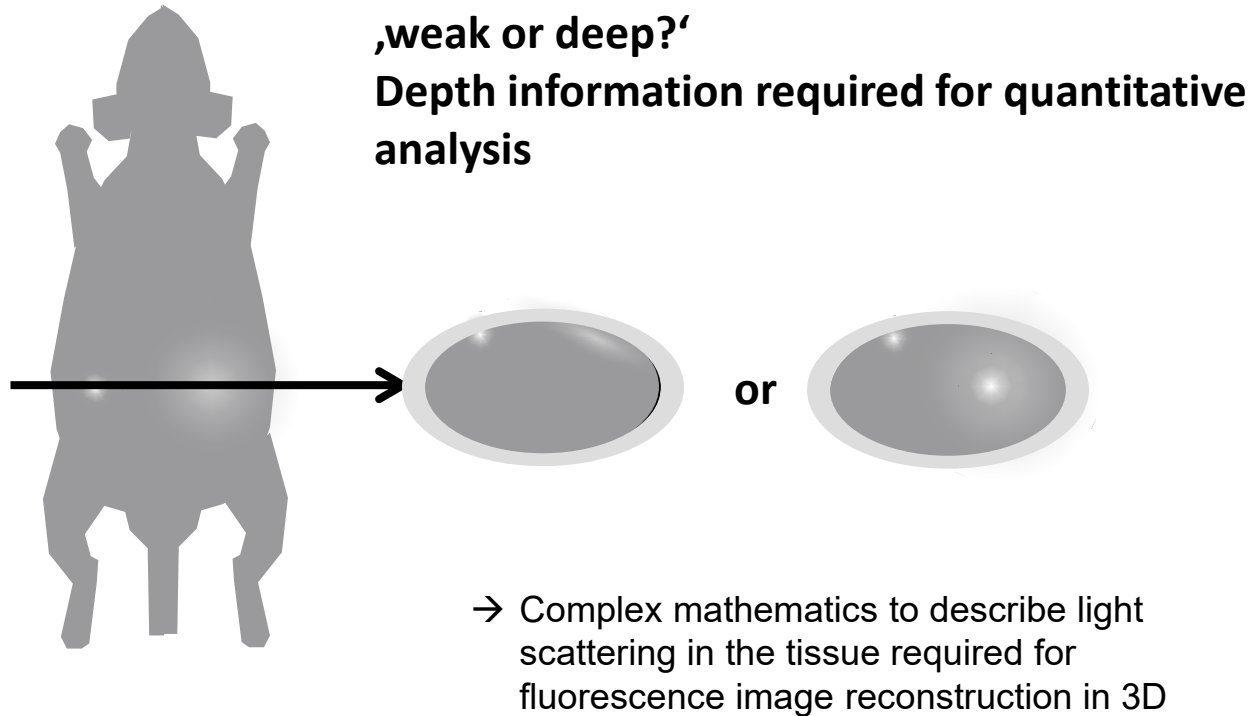
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Fluorescence Reflectance Imaging – equipment



Tung et al., Cancer Res (2000)

Development fluorescence molecular tomography (FMT)

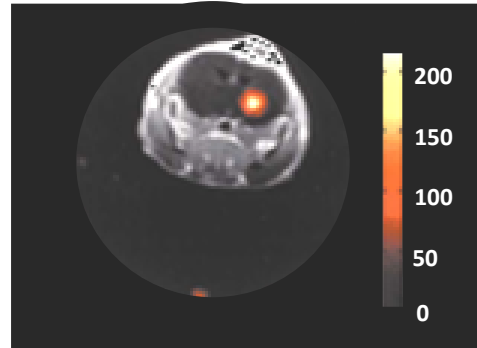


Markus Rudin, Molecular Imaging, Basic principles and applications in biomedical research, 3rd edition

Fluorescence molecular tomography- FMT



**Overlay
MRI - NIRF**



Brain tumor in rat

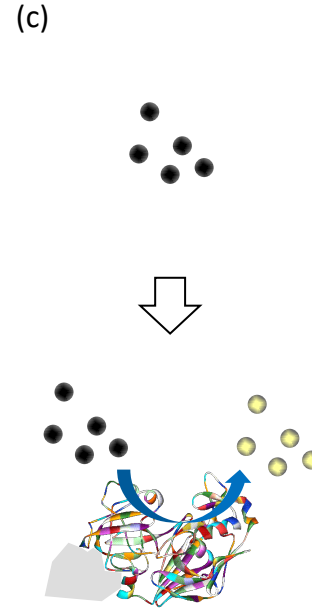
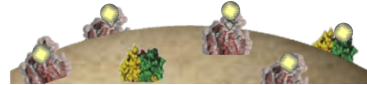
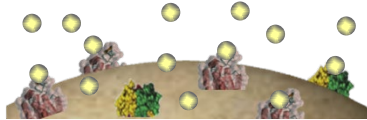
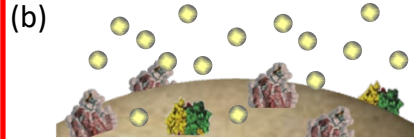
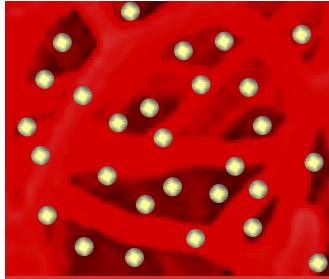
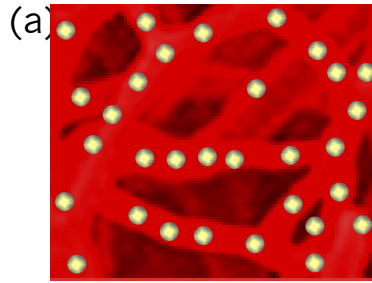
V Ntziachristos / R.Weissleder (MGH Boston)

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Contrast agents for optical imaging are fluorescent molecules: Non-specific versus target-specific contrast agents



*Fluorescent probes and fluorescent/bioluminescent
reporter proteins as contrast agents*



Intravascular imaging after FITC-dextran injection

An example of non-specific imaging

Intravenous FITC dextran (70 kDa) injection into 48h hours old zebrafish embryo to visualize the zebrafish vasculature by 2-photon microscopy

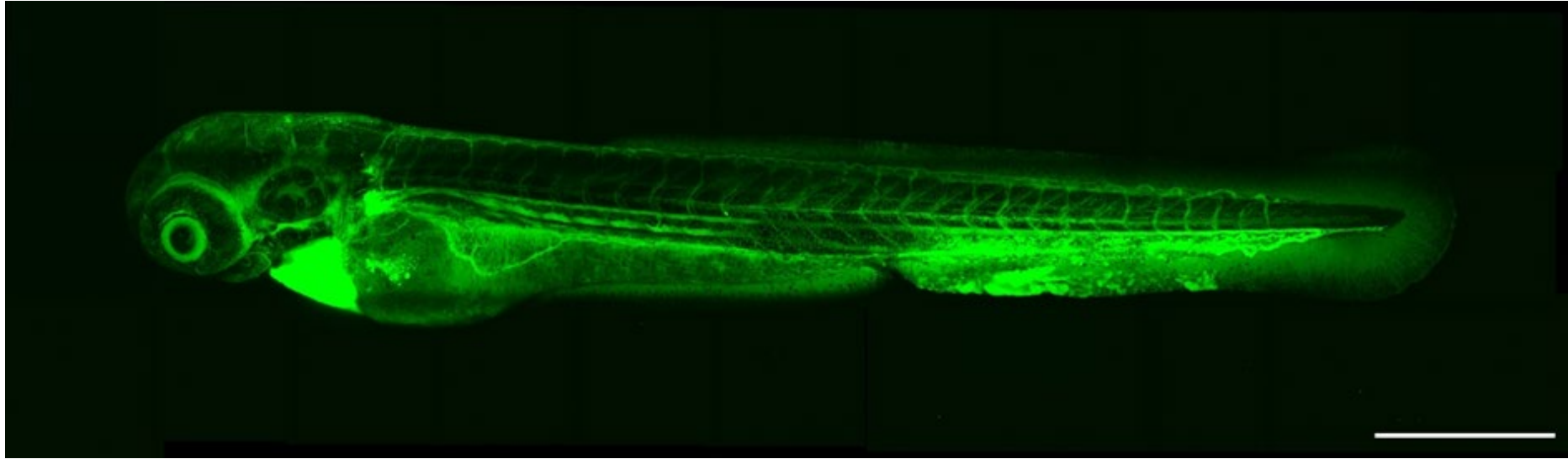


Image Courtesy, Lynn Gasser, Master student ZHAW Pharma team

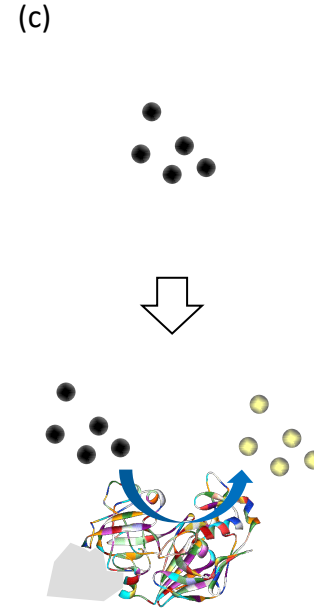
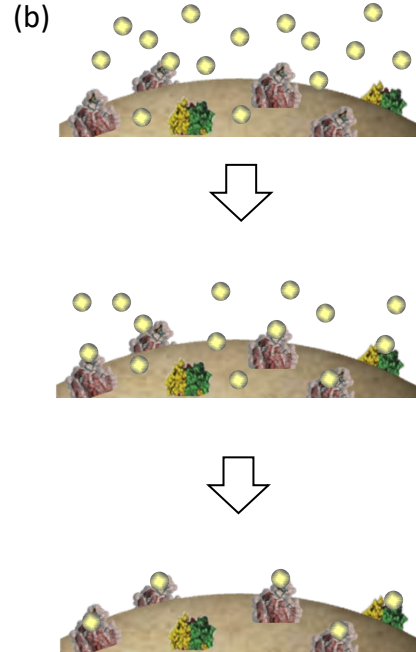
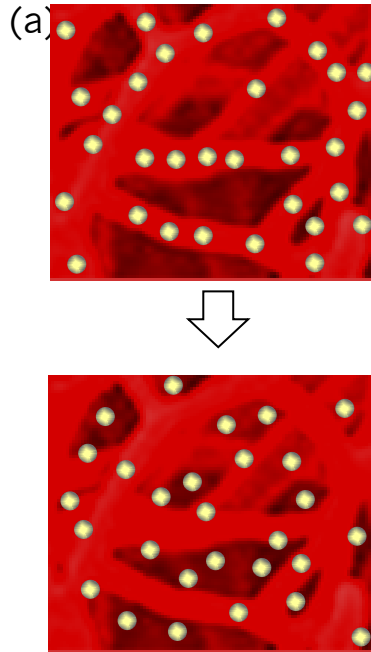
Non specific labeling of malignant tissue during surgery



Intraoperative surgery:
NIR imaging helps to
distinguish healthy from
malignant tumor tissue

Contrast agents for optical imaging are fluorescent molecules:

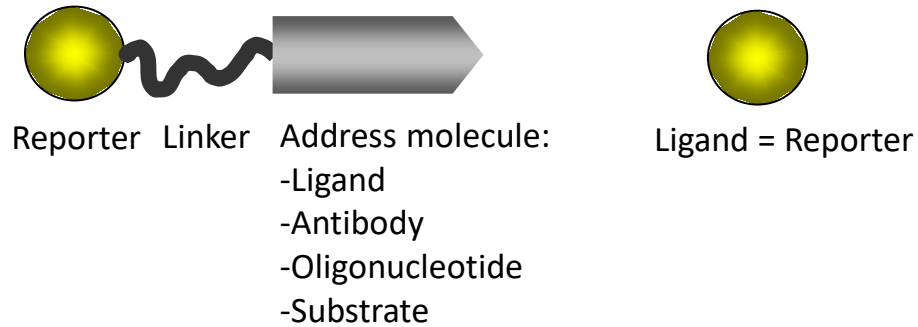
Non-specific versus target-specific contrast agents



Fluorescent probes and fluorescent/bioluminescent reporter proteins as contrast agents

Design of target specific contrast agents

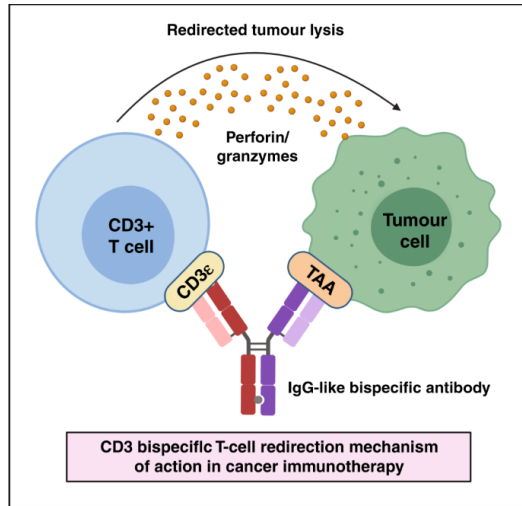
Direct targeting: Coupling of reporter to a target-specific moiety



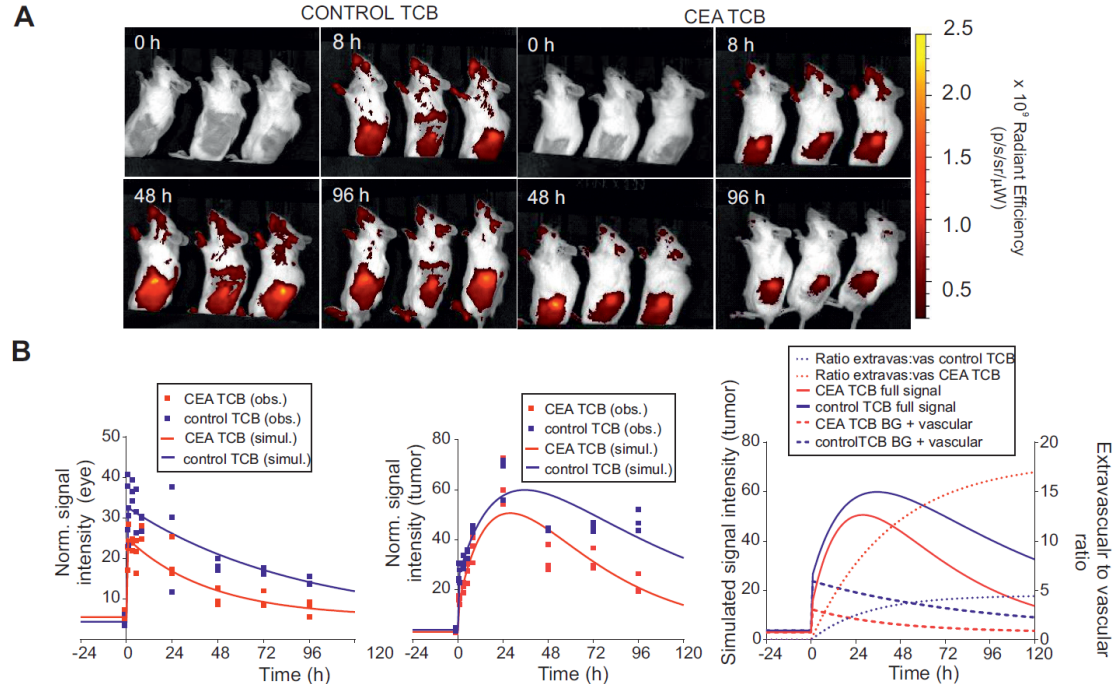
Typically, fluorescent reporter probes are either fluorescent molecules or fluorescent reporter proteins

Example: imaging cancer cells – using fluorescently labeled antibodies

Example: T-cell bispecific antibody binding Carcinoembryonic antigen (CEA), overexpressed in cancer → **fluorescently labeled with Alexa 647**



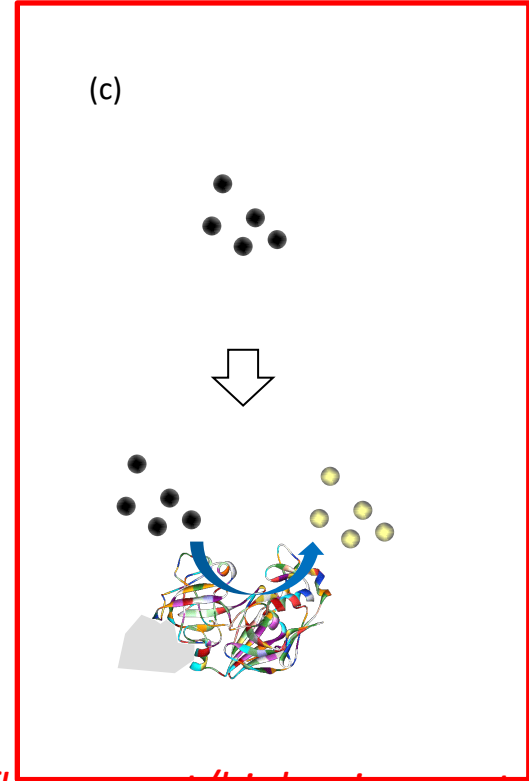
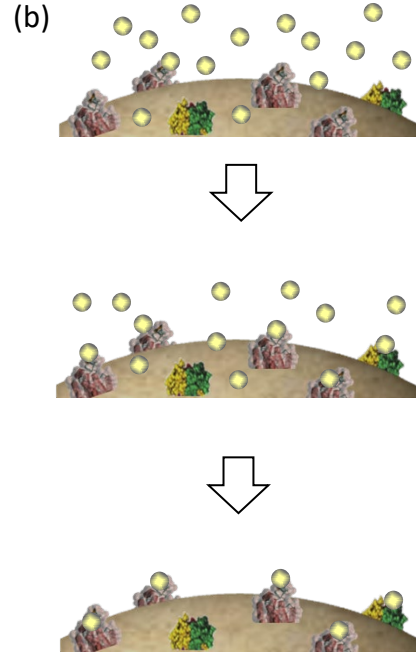
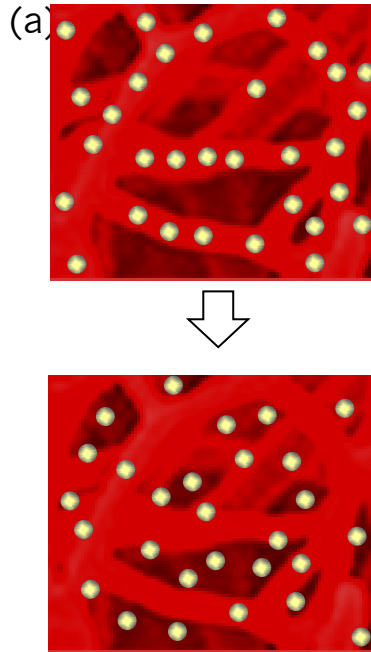
Singh, A., Dees, S. & Grewal, I.S. *Br J Cancer* **124**, 1037–1048 (2021).



Lehmann et al., *Clinical Cancer Research*, 2016

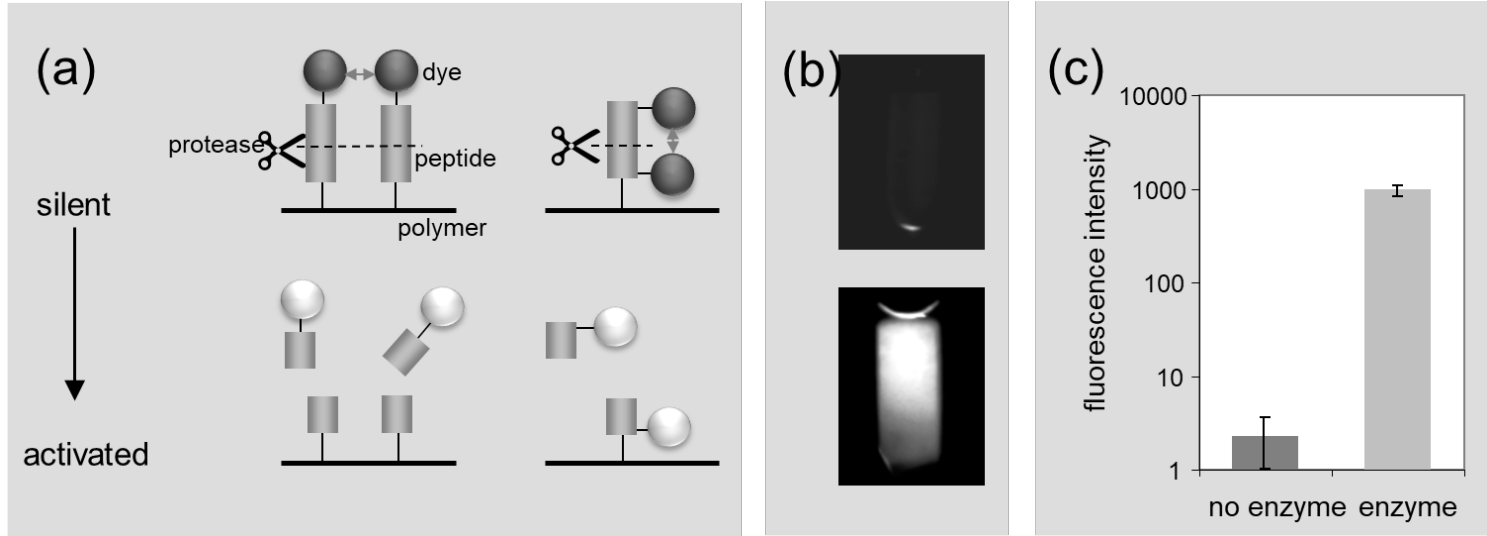
Contrast agents for optical imaging are fluorescent molecules:

Non-specific versus target-specific contrast agents



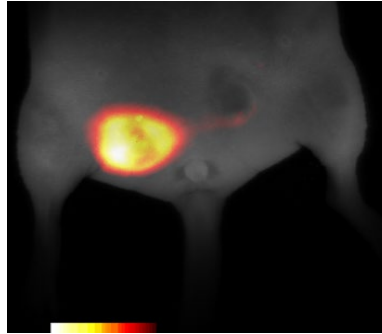
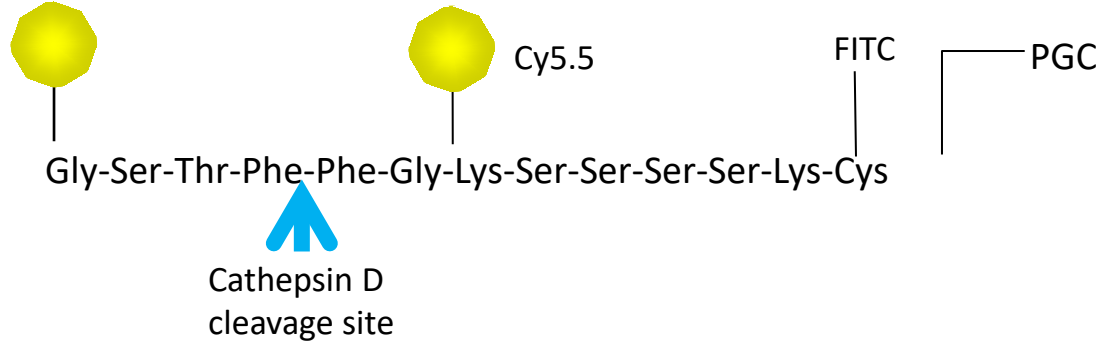
Fluorescent probes and fluorescent/bioluminescent reporter proteins as contrast agents

Design of «smart» (imaging) probes



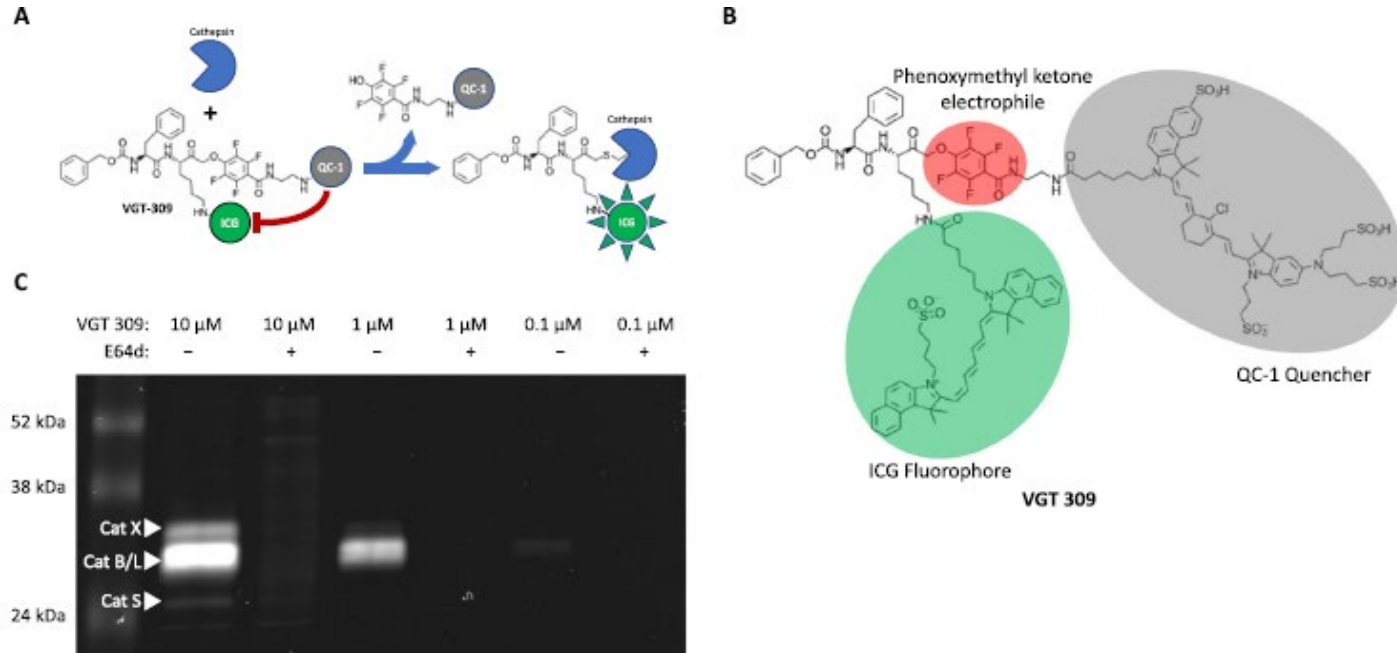
Smart probes: Quenching - dequenching

Cathepsin D specific NIRF probe

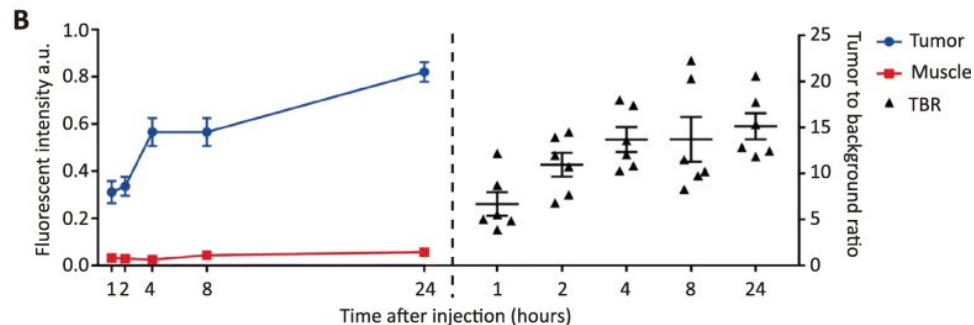
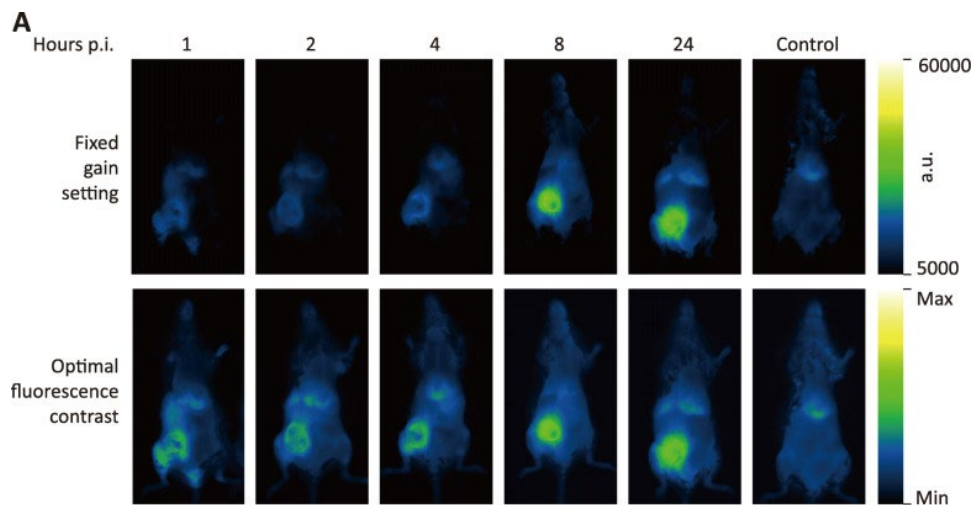


Tung et al. Cancer Res (2000)

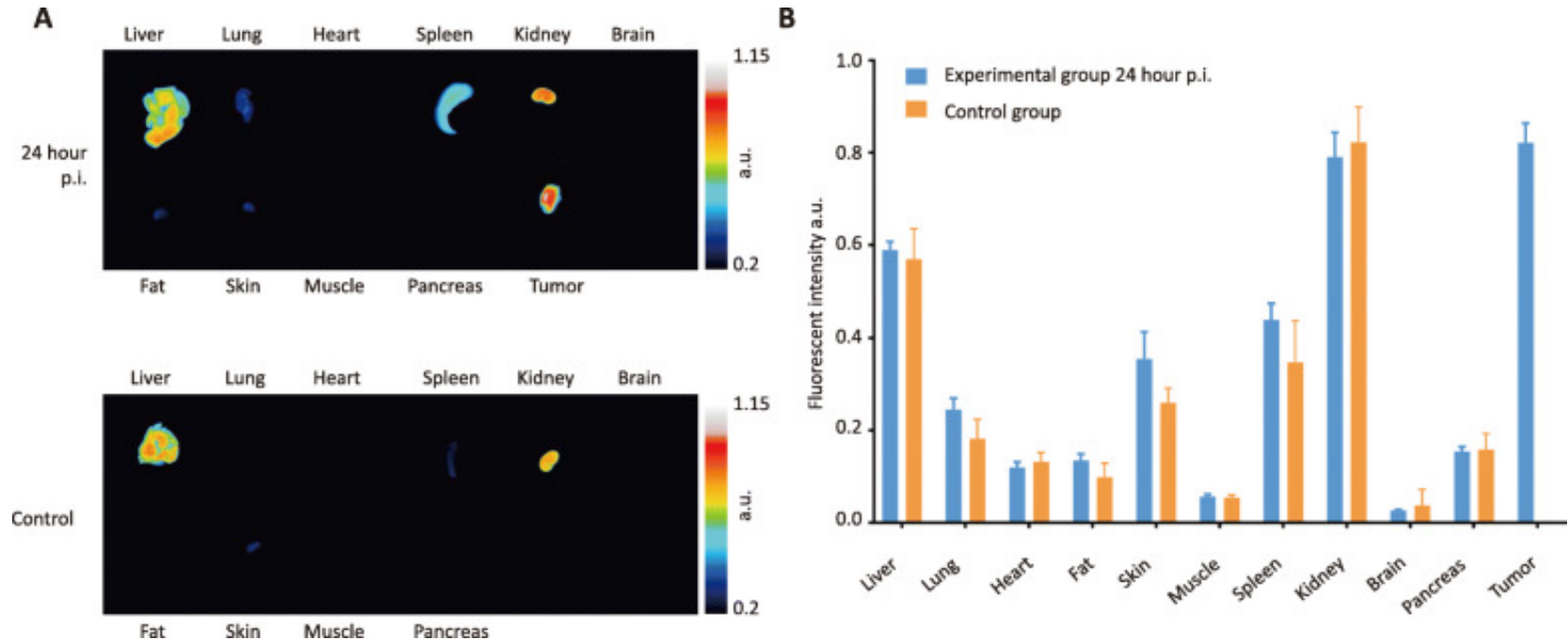
Imaging cysteine cathepsin activity – VGT309 probe



Imaging cysteine cathepsin activity in tumor bearing mice



Assessment of *in vivo* biodistribution of VGT309 in mice



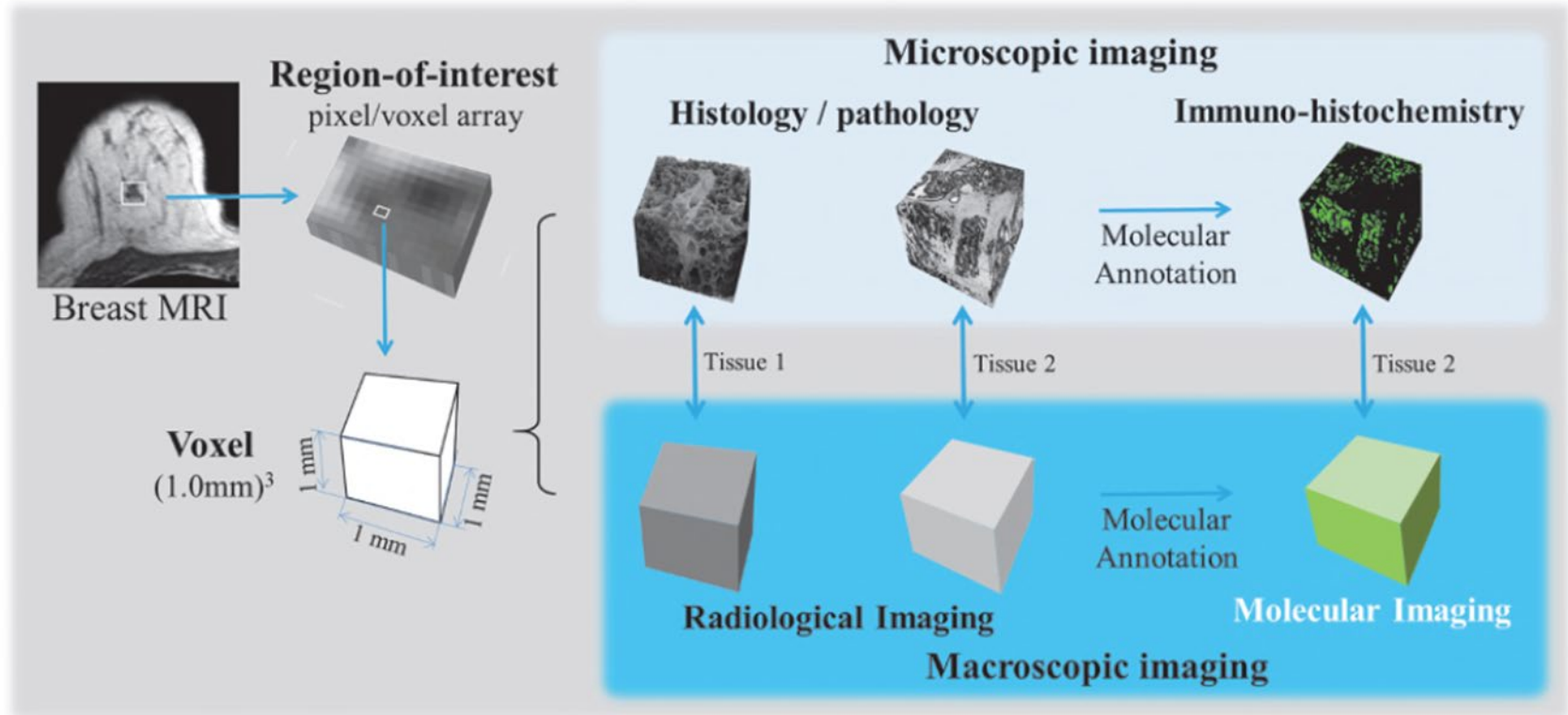
EJNMMI Res. 2020 Sep 29;10(1):111. doi: 10.1186/s13550-020-00688-0.

Program today

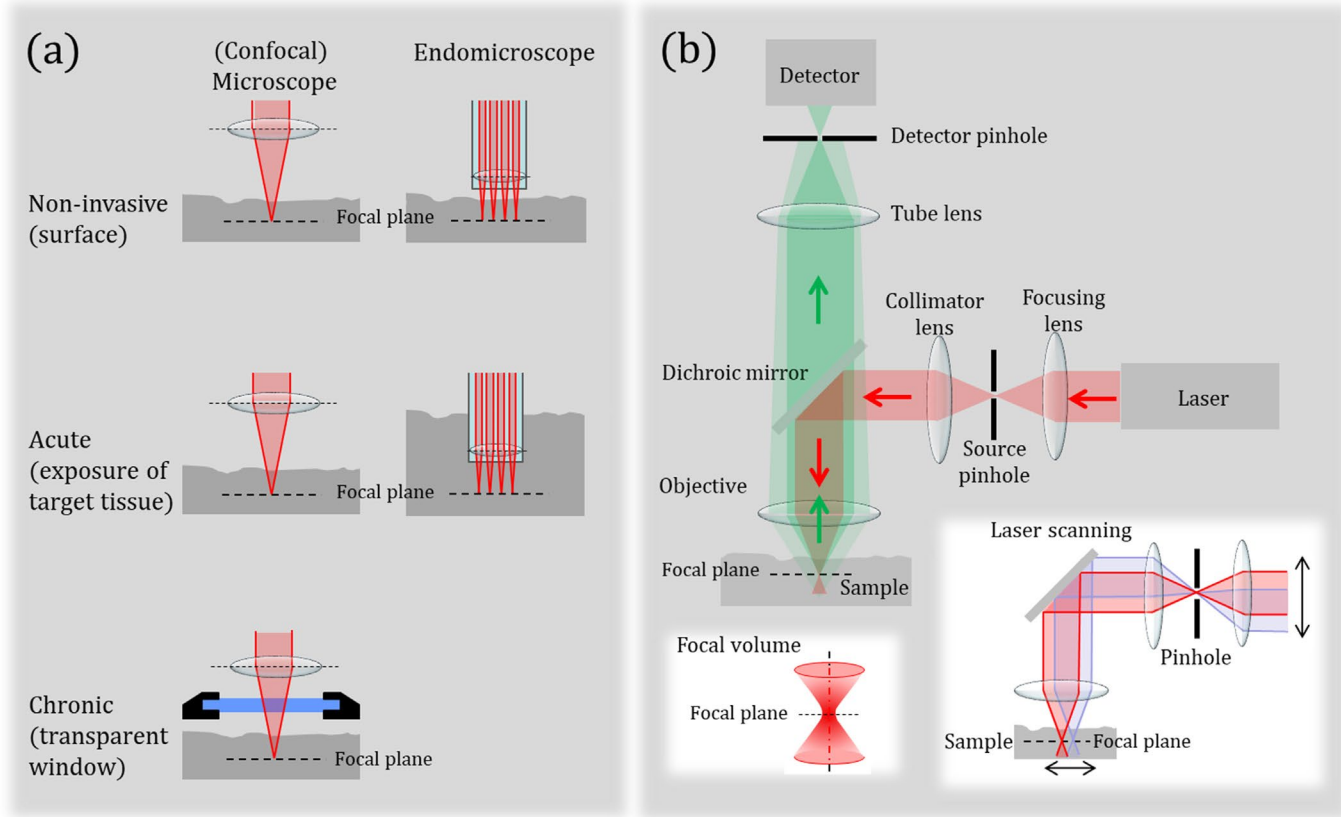
Introduction to **optical molecular imaging**:

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From macroscopic to microscopic fluorescence imaging

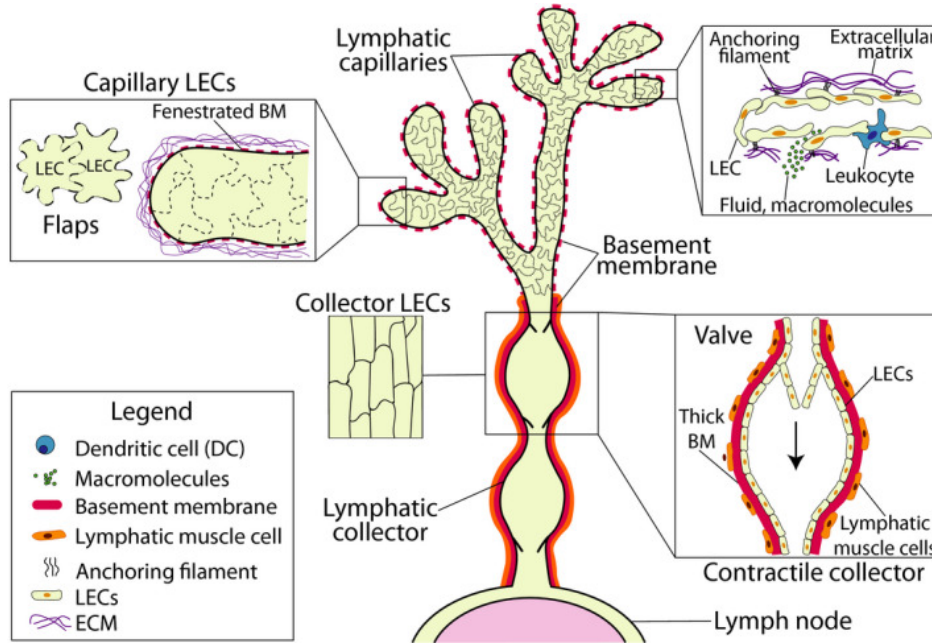


Intravital microscopy techniques: confocal microscopy



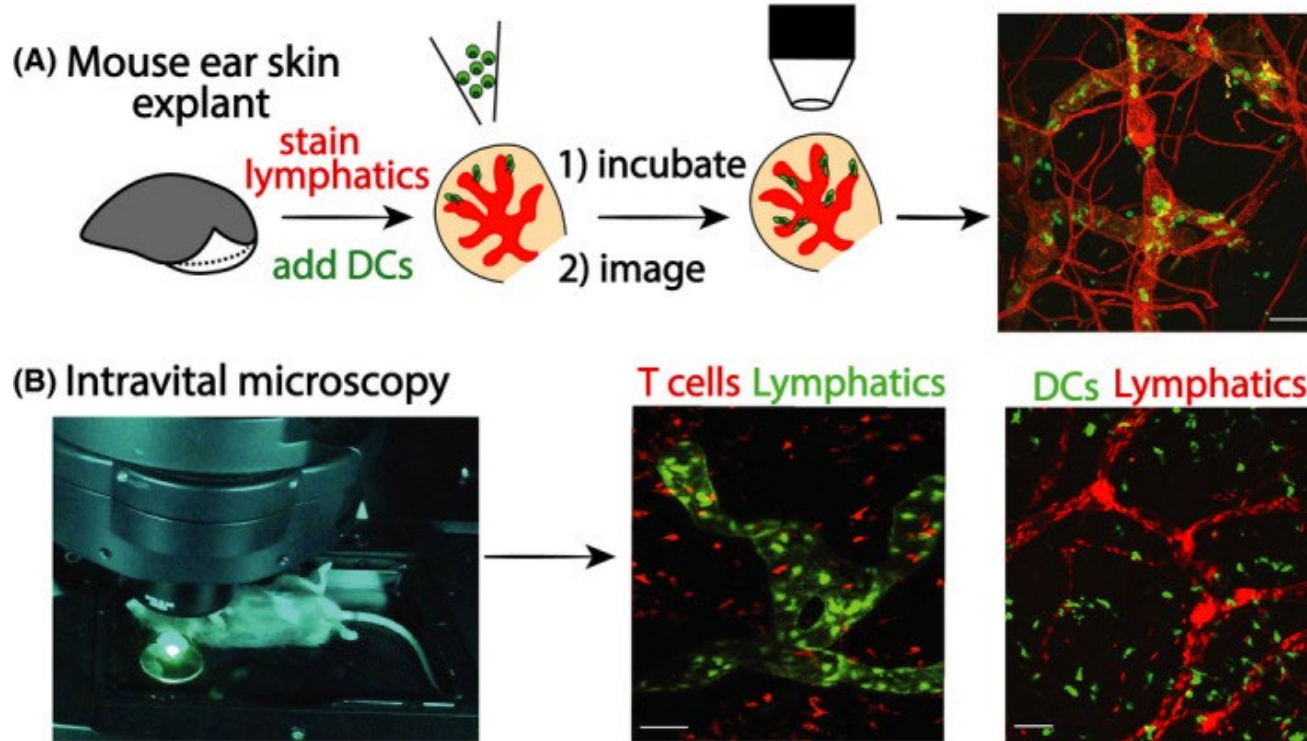
Markus Rudin, *Molecular Imaging, Basic principles and applications in biomedical research*, 3rd edition

Example: intravital confocal microscopy for imaging the trafficking of dendritic cells through afferent lymphatics

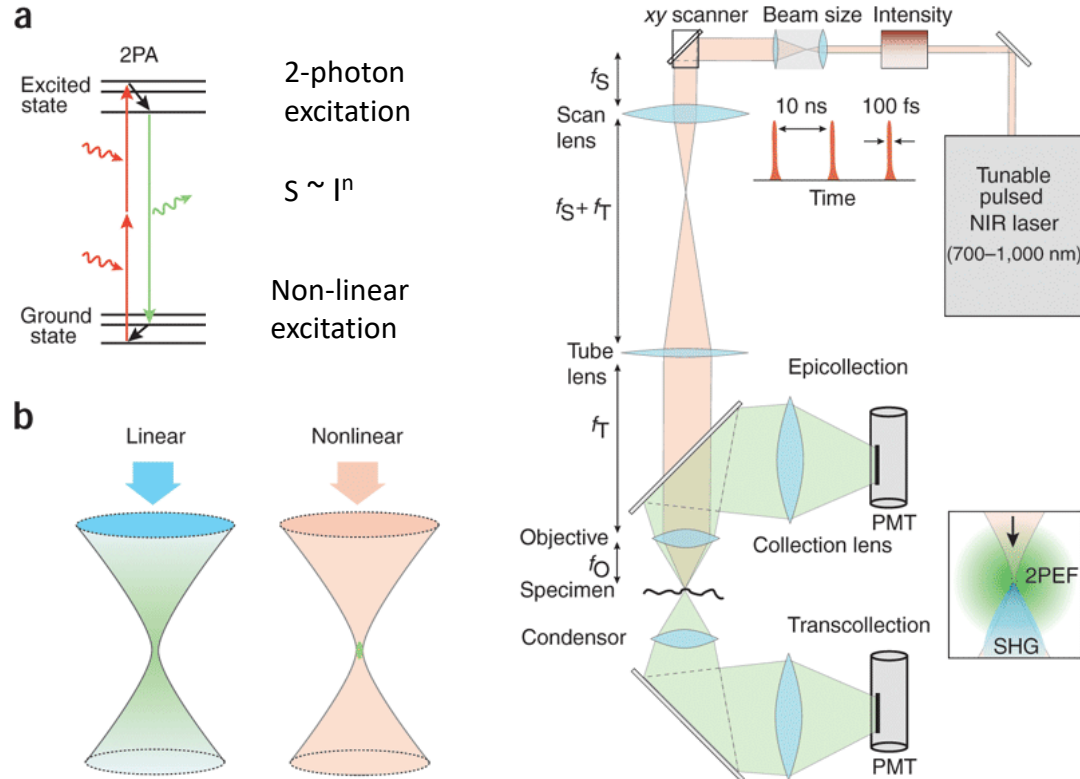


Collado-Diaz, V., Medina-Sanchez, J. D., Gkoutidi, A. O., & Halin, C. (2022). *Immunological reviews*, 306(1), 43–57.

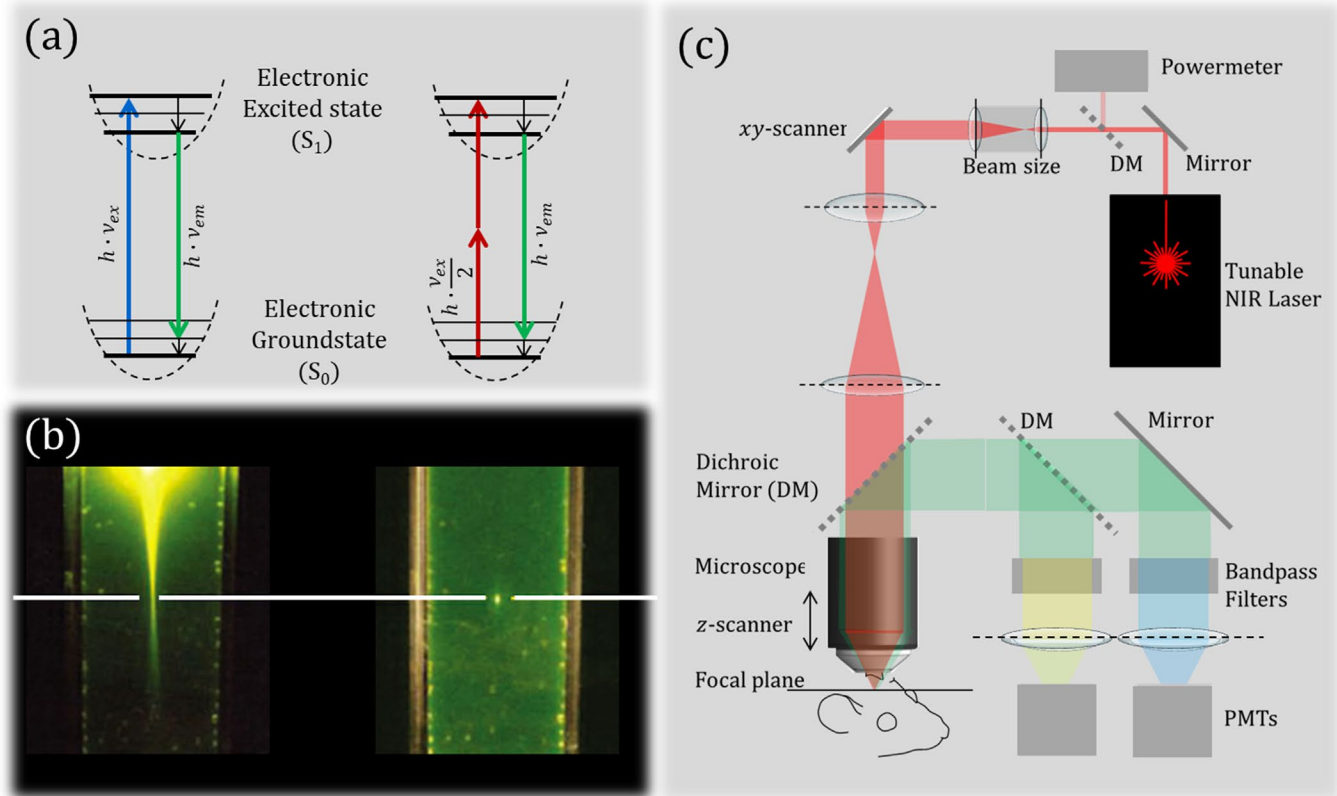
Example: intravital confocal microscopy for imaging the trafficking of dendritic cells through afferent lymphatics



Principles of 2-photon excitation- *in vivo* imaging at cellular and subcellular resolution



Principles of 2-photon excitation- *in vivo* imaging at cellular and subcellular resolution



Markus Rudin, *Molecular Imaging, Basic principles and applications in biomedical research*, 3rd edition

In vivo efficacy imaging by 2P microscopy



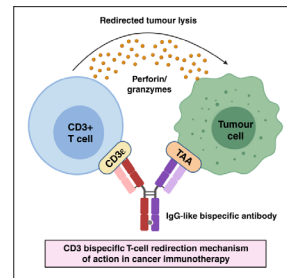
S. Alexander, Friedl Lab,
NCMLS, Nijmegen

Mouse strain: NOG (immune-deficient mice)

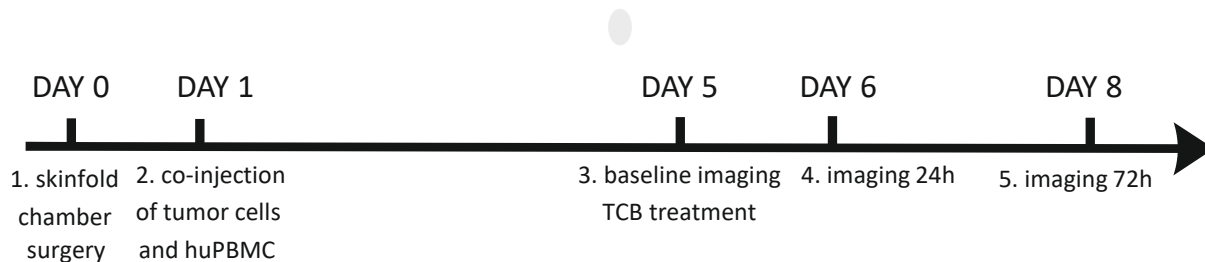
Tumor cells: 1 Mio LS174T RFP

huPBMC: 10 Mio, thereof 5 Mio CFSE labeled T-cells

→ s.c. injection of tumor/PBMC co-grafts into
skinfold chambers of mice

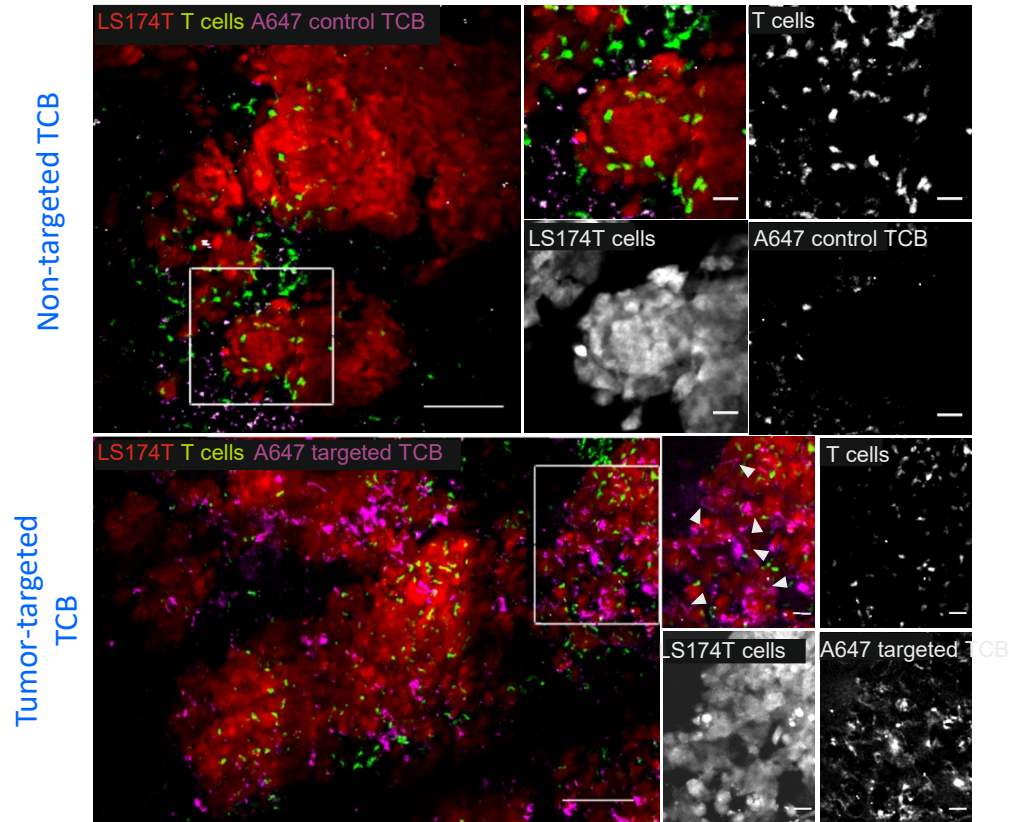


Experimental Outline



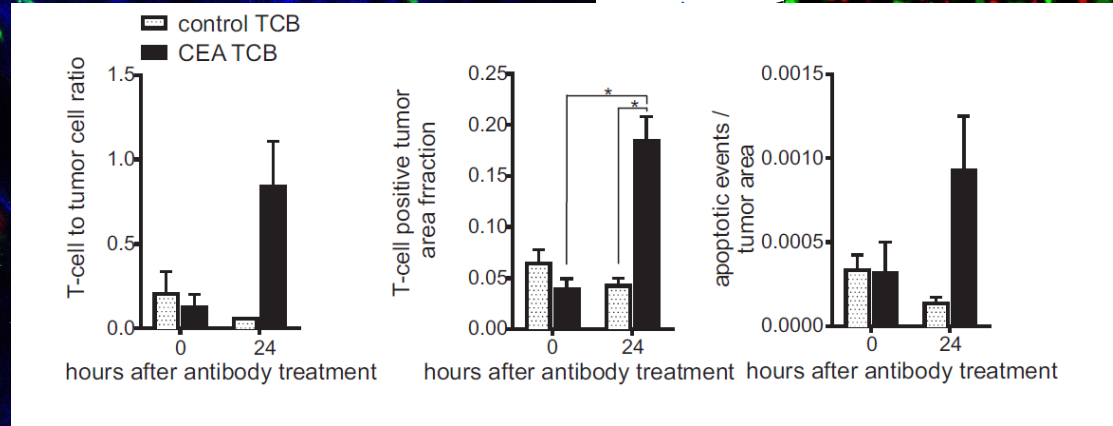
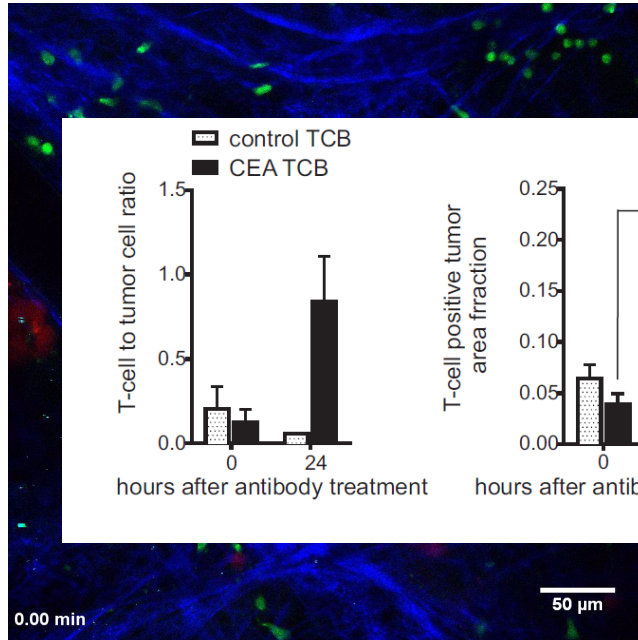
Lehmann et al., *Clinical Cancer Research*, 2016

Assessment of *in vivo* targeting by 2P imaging

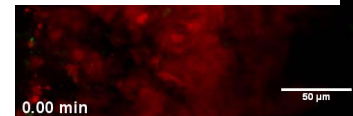
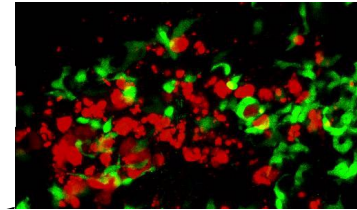


Roche TCB: mode of action 2P imaging

Baseline Imaging, 4 d after window
implantation



24 h post antibody injection



LS174T RFP T cells (CFSE) SHG collagen fibres

Program today

Introduction to **optical molecular imaging**:

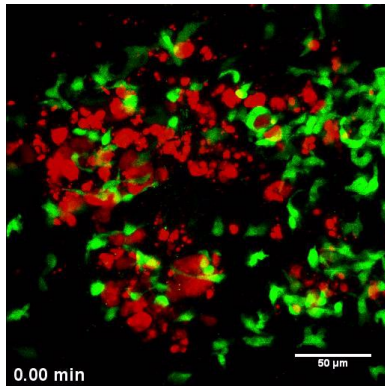
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Optical *in vivo* imaging can be employed to longitudinally track molecular and cellular processes *in vivo*

Fluorescence imaging

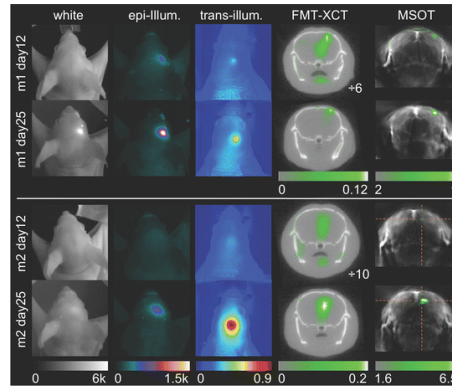
Intravital microscopy

Visualizing therapy induced killing of tumor cells by T cells



Fluorescence reflectance imaging or tomography

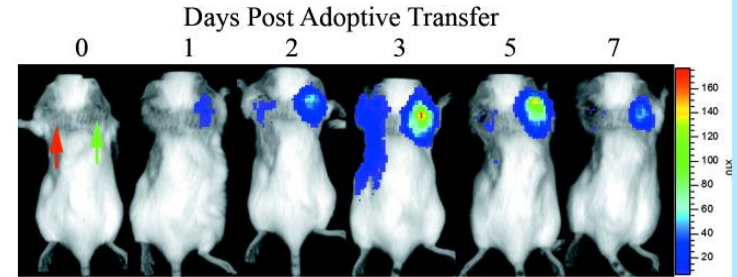
Monitoring tumor growth (glioblastoma in the mouse brain)



Deliolanis et al, Molecular Imaging and Biology, 2014

Bioluminescence imaging

Monitoring immune cell activation in a tumor model

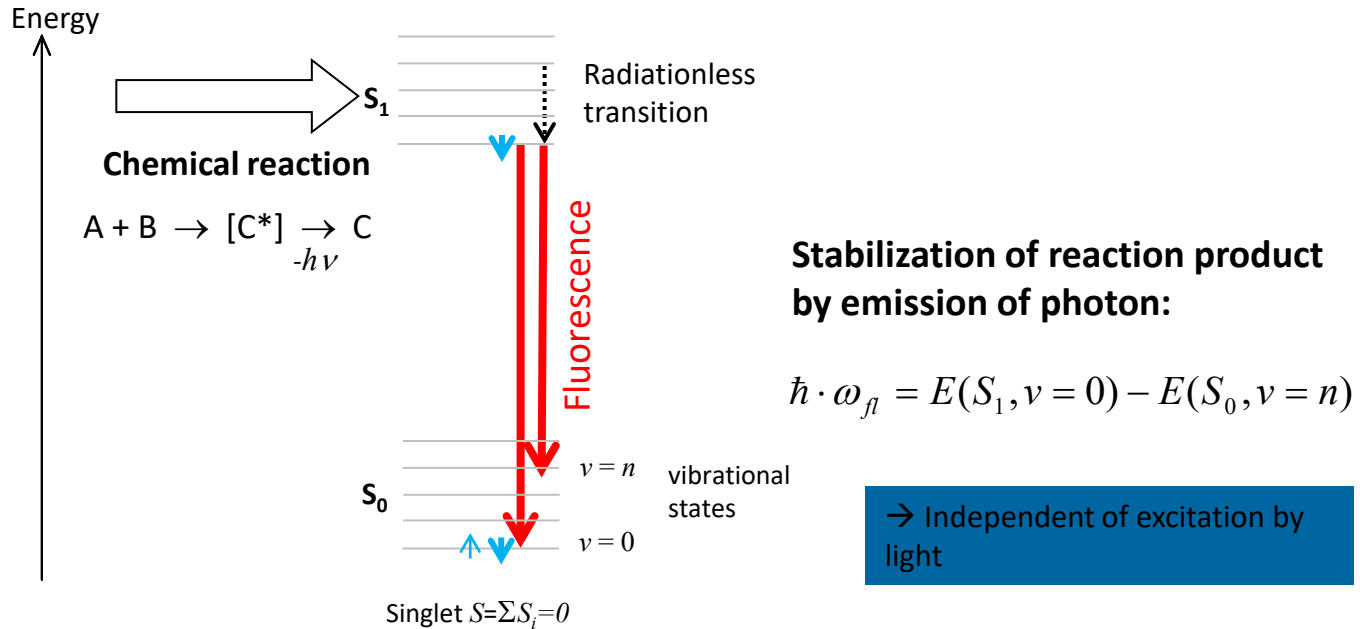


Patel et al., Cancer Research, 2010

→ Contrast generated (mostly) using exogenous fluorescent and bioluminescent probes

Bioluminescence / Chemoluminescence

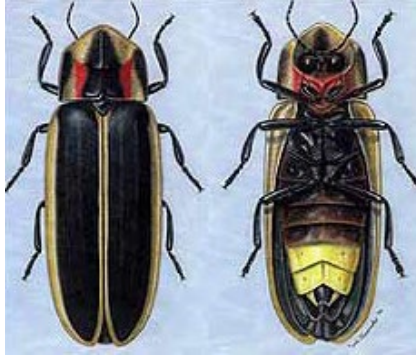
Molecular system in excited state following chemical reaction



Bioluminescence: typically enzyme catalyzed reactions

Bioluminescence / Chemoluminescence

Best known bioluminescent enzyme family: Luciferases



Photinus pyralis

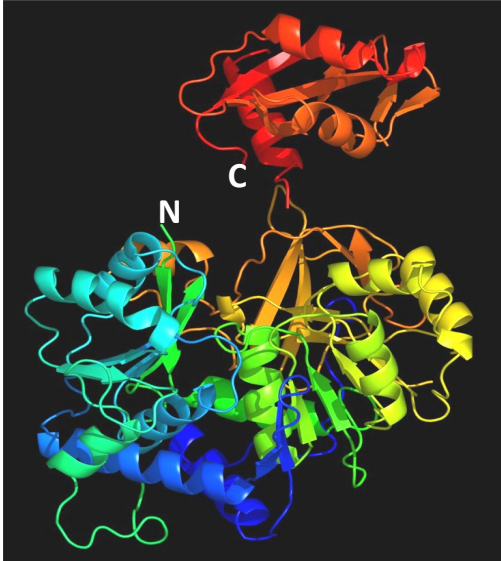


Renilla reniformis



Renilla mullerei

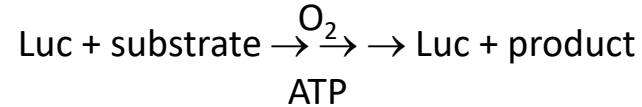
Bioluminescence



Firefly luciferase:

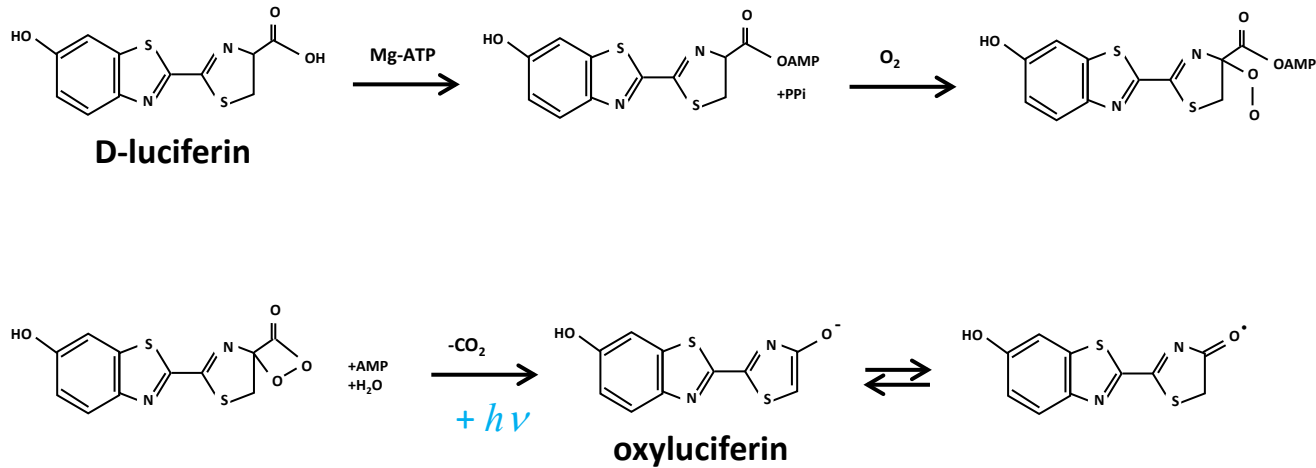
- 550 amino acid protein consisting of a globular N-terminus and a smaller C-terminus

- **Luciferases are oxygenases:**

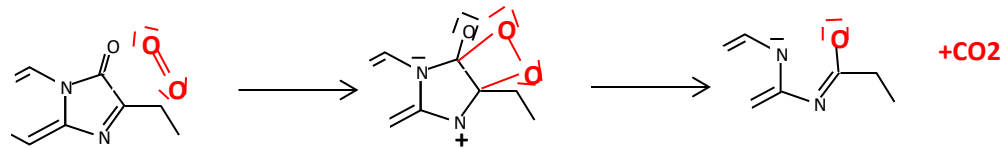
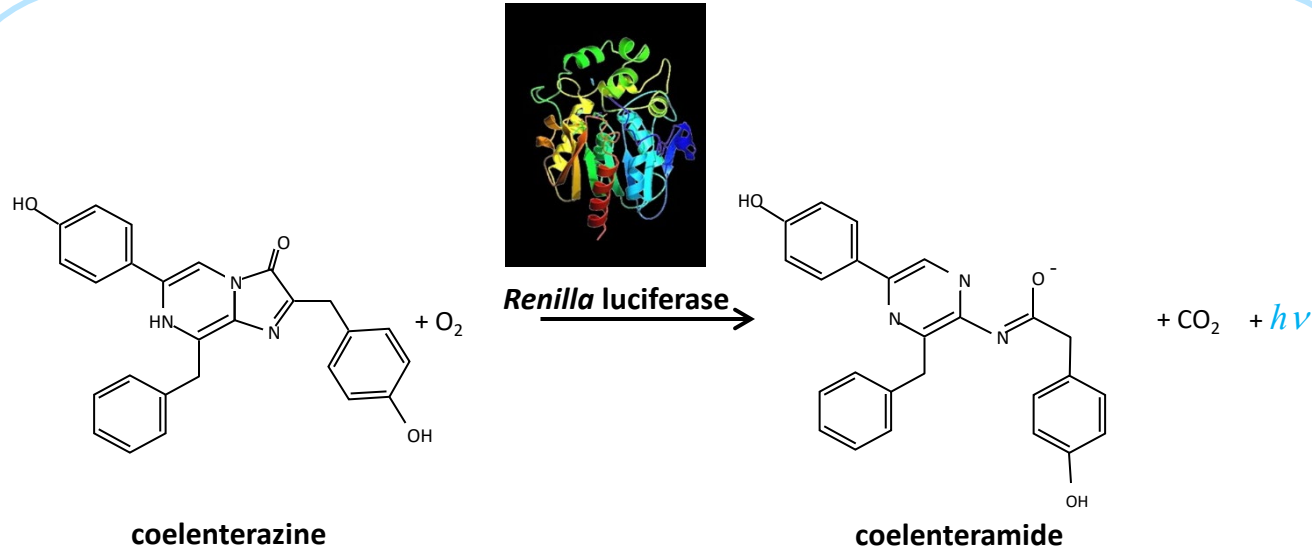


- The native enzyme produces light with $\lambda_{e,max} = 560\text{nm}$ with significant spectral contributions above 600nm
- The natural substrate is D-luciferin, D-(-)-2-(6'-hydroxy-2-benzothiazolyl)-thiazone-4-carboxylic acid

Bioluminescence: Oxidation of D-luciferin

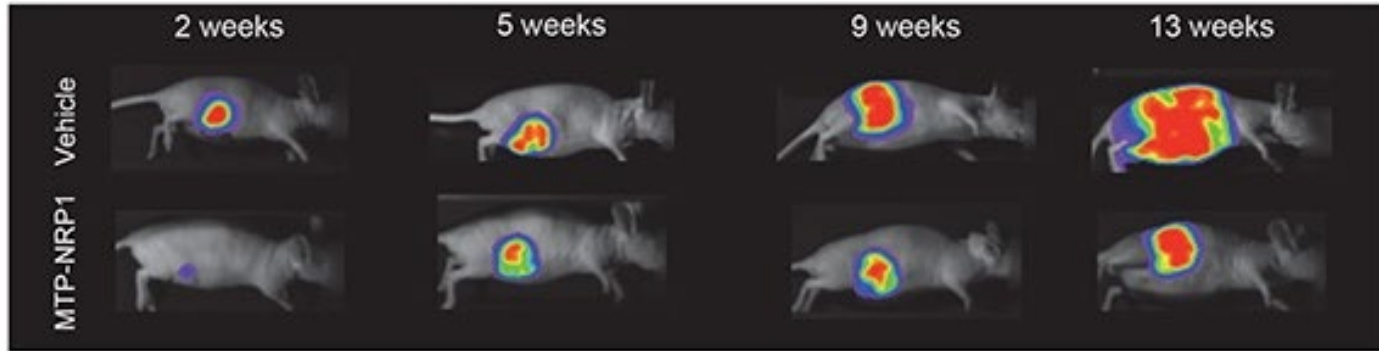


Renilla luciferase: Oxidation of Coelenterazine



Renilla luciferase catalyzed oxygenation of coelenterazine to coelenteramide

Imaging cancer cells – luciferase as reporter gene



[*Oncotarget*. 2016 Aug 23; 7\(34\): 54723–54732.](#)

Imaging cancer cells using reporter genes:

- + longitudinal studies
animals / cells required
- + fewer animals

- + multimodal reporters available

- genetic modification of

- not full translational,
mainly preclinical tool

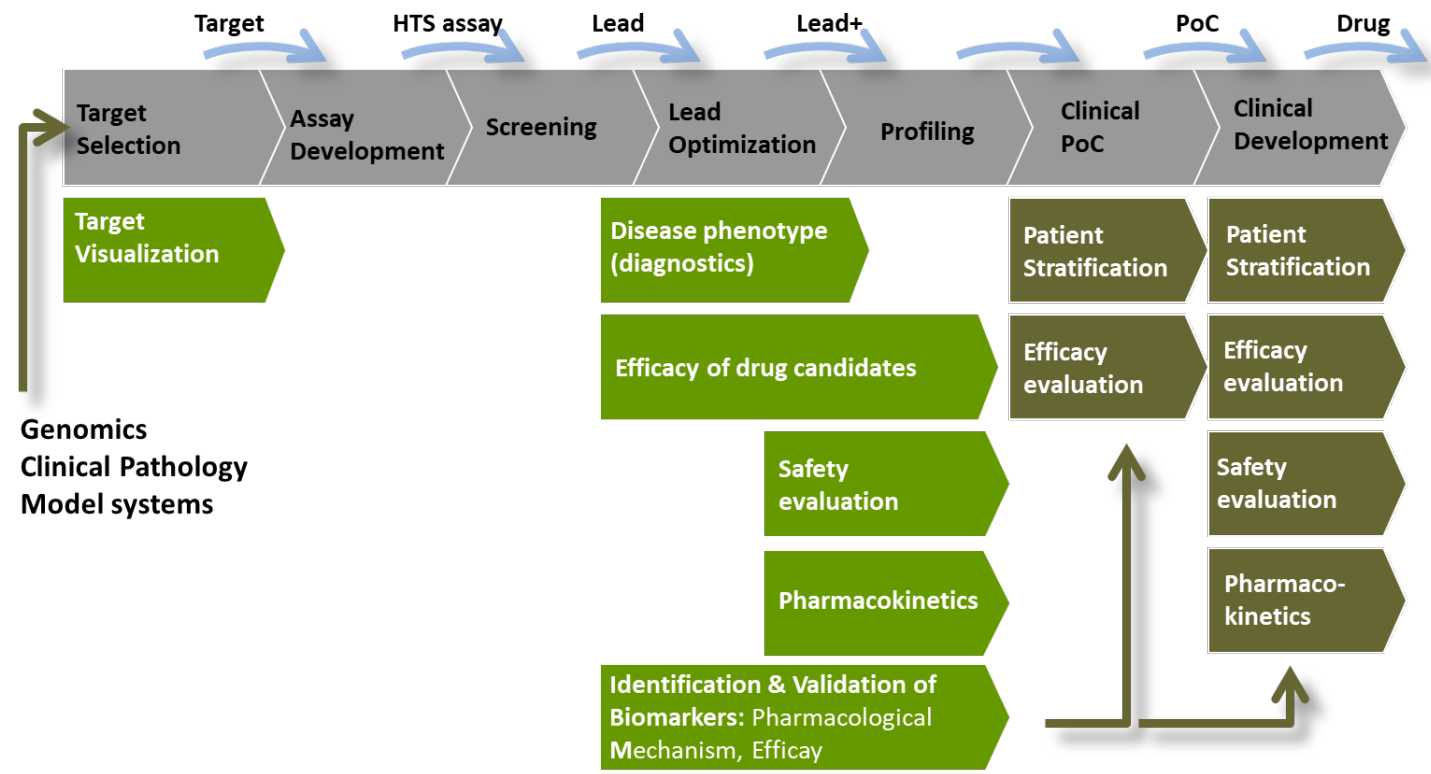
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Optical imaging methods are used for diagnostic purposes, for imaging therapeutic efficacy, but also for imaging in the preclinical drug development



Target Validation: Disease Relevance, 'Druggability'

Optical *in vivo* imaging

Optical *in vivo* imaging – why???

- Cost- and time efficient
- Extremely versatile (exploiting a wide range of light-tissue interactions) to generate contrast
- An increasing number of reporter probes available which allow assessment of gene expression, protein function, protein-protein interactions and other cellular processes
- Microscopic and macroscopic imaging of the same probes allows evaluation of processes detected by whole-body imaging at the cellular level
- Can be combined with other imaging modalities (e.g. with MRI/PET)

Optical *in vivo* imaging

Optical *in vivo* imaging – why not???

- Limited by tissue penetration depth
- Full quantification difficult
- For macroscopic (whole-body) imaging: spatial resolution in mm range

Summary- take home messages

- ***In vivo* optical imaging is a cost-effective, versatile tool** to assess cellular and molecular processes in a spatio-temporo controlled manner, primarily employed for preclinical imaging purposes although there are also clinical applications
- Fluorescence optical imaging comprises **macroscopic (reflectance, tomography) and microscopic imaging readouts**
- Fluorescence **multiphoton microscopy allows for deep tissue imaging** at cellular and subcellular resolution
- **Bioluminescence imaging** is a macroscopic imaging readout for **high sensitivity** imaging of luciferase as reporter gene
- **Imaging contrast agents** used for optical imaging are either **fluorescent dyes** or **genetically-encoded proteins** (meanwhile there is a tool box of fluorescent proteins available covering a wide range of the visible spectrum)
- Fluorescence imaging agents may be **targeted or non-targeted and can be activatable (smart probes or not)**

Thank you for your attention!!

leht@zhaw.ch